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2. Course Objectives

- To explain the fundamentals of data science
- To describe and understand data for analysis
- To describe and analyze relationships between variables.
- To apply various data pre-processing strategies
- To apply data wrangling on data for further processing



3. Pre-Requisites

Semester-III

Foundations Of Data
Science

Semester-II

Java Programming

Semester-I

C Programming

4.SYLLABUS

22AI402

FOUNDATIONS OF DATA SCIENCE

L T P C

2 0 2 3

UNIT I INTRODUCTION

6+6

Data Science: Benefits and uses – facets of data - Data Science Process: Overview – Defining research goals – Retrieving data – data preparation - Exploratory Data analysis – build the model – presenting findings and building applications - Data Mining - Data Warehousing – Basic statistical descriptions of Data.

List of Exercise/Experiments:

1. Explore a dataset and describe it using basic statistics and represent it using simple graphs.
2. Group data and find averages or counts for each group.
3. Build a simple machine learning model from a dataset and show how well it works.

UNIT II DESCRIBING DATA

6+6

Types of Data - Types of Variables -Describing Data with Tables and Graphs – Describing Data with Averages - Describing Variability - Normal Distributions and Standard (z) Scores.

List of Exercise/Experiments:

1. Use graphs to find patterns or trends in a dataset.
2. Compare two or more columns to find relationships in a dataset.
3. Find the average, spread, and z-scores for numbers in a dataset.
4. Find which values are close to the average and which are far using z-scores.

UNIT III DESCRIBING RELATIONSHIPS

6+6

Correlation –Scatter plots –correlation coefficient for quantitative data –computational formula for correlation coefficient –

Regression –regression line –least squares regression line –

Standard error of estimate – interpretation of r^2 –multiple regression equations – regression towards the mean.

List of Exercise/Experiments:

1. Use a scatter plot to see how two variables are related. Calculate the correlation coefficient to measure the strength of the relationship.
2. Find the line of best fit (regression line) for two variables. Calculate r^2 to see how well the line fits the data.

3. Use a Multiple Linear Regression model to predict a target variable from multiple inputs, and plot the predicted vs. actual values to evaluate model performance.
4. Create a dashboard or summary using charts and key numbers.

UNIT IV DATA PREPROCESSING 6+6

Data Preprocessing – Purpose – Tools – Levels – Cleaning up the Table – Unpacking – Restructuring – Reformulating the Table - Data Cleaning: – Missing Values – Outliers – Errors - Data Fusion vs Data Integration – Directions – Adding attributes – data objects - Entity identification – Data Reduction vs data redundancy – Types – Performing Numerosity Data reduction – Sampling – Principal Component Analysis – Data Transformation – Normalization and Standardization – Discretization – Smoothing – Aggregation - Binning.

List of Exercise/Experiments:

1. Clean and preprocess a dataset by handling missing values, scaling features, treating outliers, and creating dummy variables.
2. Train and tune a model using cross-validation to understand and manage the bias-variance trade-off.
3. Perform Outlier Analysis on any data set.

UNIT V DATA WRANGLING

6+6

Processing Uni-dimensional Data: Creating Vectors - Inspecting the Data Distribution with Histograms- Aggregating Numerical Data - Arithmetic Operators - Indexing vectors- Multidimensional data: Creating matrices - Reshaping - visualization (2d,3d) - Heterogeneous Data: Data frames - Aggregation - Transformation - Indexing - Accessing Database: Filtering - Ordering - Removing Duplicates - Grouping and Aggregating - Joining - Handling with Many Files..

List of Exercise/Experiments:

1. Create and reshape matrices, apply matrix operations, and visualize data using 2D heatmaps and 3D surface plots.
2. Build a DataFrame, perform aggregation and transformation, filter and sort data, remove duplicates, group, and join tables.
3. Simulate SQL-style operations like filtering, ordering, deduplication, and grouping in a DataFrame.

TOTAL:30+30=60 PERIODS

SYLLABUS

TEXTBOOKS:

1. Robert S. Witte and John S. Witte, "Statistics", Eleventh Edition, Wiley Publications, 2017.

(Units 2, 3)

2. Jacqueline Kazil and Katharine Jarmul, Data Wrangling with Python: Tips and Tools to Make Your Life Easier, O'Reilly Media, 1st Edition, 2016.

REFERENCES:

1. David Cielen, Arno D. B. Meysman, and Mohamed Ali, "Introducing Data Science", Manning

Publications, 2016. (Unit 1)

2. Roy Jafari, Hands-On Data Preprocessing in Python: Learn how to effectively prepare data for successful data analytics, Packt Publications, First Edition, 2022. (Unit 4)

3. Marek Gagolewski, Minimalist Data Wrangling with Python, Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (CC BY-NC-ND 4.0), 2022. (Unit 5)

4. Hui Lin, Ming Li, Practitioner's Guide to Data Science, 1st Edition, Chapman and Hall/CRC, 2023.

5. Jiawei Han, Micheline Kamber, Jian Pei, "Data Mining: Concepts and Techniques", 3rd Edition, Morgan Kaufmann, 2012.

6. Suresh Kumar Mukhiya, Usman Ahmed, Hands-On Exploratory Data Analysis with Python: Perform EDA techniques to understand, summarize, and investigate your data, Packt Publications, 2020. Robert H. Shumway, David S. Stoffer, Time Series Analysis and Its Applications With R Examples, Fourth Edition, 2016, Springer.

7. Dr. Tirthajyoti Sarkar, Shubhadeep Roychowdhury, Data Wrangling with Python: Creating actionable data from raw sources, Packt Publications, First Edition, 2019.

8. Robert H. Shumway, David S. Stoffer, Time Series Analysis and Its Applications With R Examples, Fourth Edition, 2016, Springer.

9. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements Of Statistical Learning: Data Mining, Inference, and Prediction, Springer, Second Edition, 2017.

10. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, O'Reilly, Third Edition, 2022.

11. Jake VanderPlas, "Python Data Science Handbook – Essential tools for working with data", O'Reilly, 2017.

12. Ashwin Pajankar, Aditya Joshi, Hands-on Machine Learning with Python: Implement Neural Network Solutions with Scikit-learn and PyTorch, Apress, 2022.

13. NPTEL Courses:

- a. Data Science for Engineers - https://onlinecourses.nptel.ac.in/noc23_cs17/preview
- b. Python for Data Science - https://onlinecourses.nptel.ac.in/noc23_cs21/preview

LIST OF EQUIPMENTS:

Systems with Anaconda, Jupyter Notebook, Python, Pandas, NumPy, Mathplotlib. NPTEL Courses:

Data Science for Engineers - https://onlinecourses.nptel.ac.in/noc23_cs17/preview

Python for Data Science - https://onlinecourses.nptel.ac.in/noc23_cs21/preview

LIST OF EQUIPMENTS:

Systems with Anaconda, Jupyter Notebook, Python, Pandas, NumPy, Mathplotlib

1. COURSE OUTCOMES

At the end of this course, the students will be able to:

COURSE OUTCOMES		KL
CO1	Explain the fundamentals of data science.	K2
CO2	Illustrate the basics of data for analysis	K3
CO3	Identify relationships between variables	K3
CO4	Implement various data pre-processing strategies	K4
CO5	Implement data wrangling for further processing of data..	K3
CO6	Apply Data preprocessing on real-time data sets	K4

CO – PO/PSO Mapping

1.CO – PO /PSO Mapping Matrix

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS 01	PS 02	PS 03
1	3	2	2		1			1	1	1		1	2	3	3
2	3	3	3		3			1	1	1		1	2	3	3
3	3	3	3		3			3	3	3		3	2	3	3
4	3	3	3		3			3	3	3		3	2	3	3
5	3	3	3		3			3	3	3		3	2	3	3



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Lecture Plan

Unit - I

Lecture Plan – Unit 1- INTRODUCTION

Sl. No.	Topic	Number of Periods	Proposed Date	Actual Lecture Date	CO	Taxonomy Level	Mode of Delivery
1	Data Science	1			CO1	K2	PPT / Chalk & Talk
2	Benefits and uses	1			CO1	K2	PPT / Chalk & Talk
3	Facets of data	1			CO1	K2	PPT / Chalk & Talk
4	Data Science Process: Overview	1			CO1	K2	PPT / Chalk & Talk
5	Defining research goals	1			CO1	K2	PPT / Chalk & Talk
6	Retrieving data	1			CO1	K2	PPT / Chalk & Talk
7	Data preparation	1			CO1	K2	PPT / Chalk & Talk
8	Exploratory Data analysis	1			CO1	K2	PPT / Chalk & Talk
9	Build the model	1			CO1	K2	PPT / Chalk & Talk

10	Presenting findings and building applications	1			CO1	K2	PPT / Chalk & Talk
11	Data Mining	1			CO1	K2	PPT / Chalk & Talk
12	Data Warehousing	1			CO1	K2	PPT / Chalk & Talk

8. ACTIVITY BASED LEARNING

Activity name:

Building a Predictive Analytics Model for Real Time applications.

Students will have better understand about how Predictive Analysis performed for various applications by using the following steps.

- Defining Business Objectives. The project starts with using a well-defined business objective.
- Preparing Data. You'll use historical data to train your model.
- Sampling Your Data.
- Building the Model.
- Deploying the Model.

Guidelines to do an activity :

- 1) Students can form group. (3 students / team)
- 2) Identify an Organization.
- 3) Apply procedure of Data Science Process. (Follow above mentioned steps)
- 4) Conduct Peer review. (each team will be reviewed by all other teams and mentors)

UNIT-1

INTRODUCTION

9. LECTURE NOTES

1. Big Data

Big data is a collection of data sets so large or complex that it becomes difficult to process them using traditional data management techniques such as, for example, the RDBMS (relational database management systems).

1.1 Characteristics of big data

The characteristics of big data are often referred to as the three Vs:

- **Volume**—How much data is there?
- **Variety**—How diverse are different types of data?
- **Velocity**—At what speed is new data generated?
- **Veracity**: How accurate is the data?

These four properties make big data different from the data found in traditional data management tools. Consequently, the challenges they bring can be felt in almost every aspect: data capture, curation, storage, search, sharing, transfer, and visualization. In addition, big data calls for specialized techniques to extract the insights.

2. Data Science

As the amount of data continues to grow, the need to leverage it becomes more important. Data science involves using methods to analyze massive amounts of data and extract the knowledge it contains. Data science and big data evolved from statistics and traditional data management but are now considered to be distinct disciplines. Data science is an evolutionary extension of statistics capable of dealing

with the massive amounts of data produced today. It adds methods from computer science to the repertoire of statistics.

2.1 Benefits and uses of data science

Data science and big data are used almost everywhere in both commercial and noncommercial settings.

2.1.1 Commercial companies in almost every industry use data science and big data to gain insights into their customers, processes, staff, completion, and products.

- Many companies use data science to offer customers a better user experience, as well as to cross-sell, up-sell, and personalize their offerings.
- A good example of this is Google AdSense, which collects data from internet users so relevant commercial messages can be matched to the person browsing the internet.
- MaxPoint is another example of real-time personalized advertising.
- Human resource professionals use people analytics and text mining to screen candidates, monitor the mood of employees, and study informal networks among coworkers.
- Financial institutions use data science to predict stock markets, determine the risk of lending money, and learn how to attract new clients for their services.

2.1.2 Governmental organizations are also aware of data's value. Many governmental organizations not only rely on internal data scientists to discover valuable information, but also share their data with the public.

- This data is used to gain insights or build data-driven applications.

- Data.gov is the home of the US Government's open data. A data scientist in a governmental organization gets to work on diverse projects such as detecting fraud and other criminal activity or optimizing project funding.

The American National Security Agency and the British Government Communications Headquarters use data science and big data to monitor millions of individuals.

- These organizations collected 5 billion data records from widespread applications such as Google Maps, Angry Birds, email, and text messages, among many other data sources. Then they applied data science techniques to distill information.

2.1.3 Nongovernmental organizations (NGOs) use data to raise money and defend their causes.

- The World Wildlife Fund (WWF) employs data scientists to increase the effectiveness of their fundraising efforts.
- Many data scientists devote part of their time to helping NGOs, because NGOs often lack the resources to collect data and employ data scientists.
- DataKind is one such data scientist group that devotes its time to the benefit of mankind.

2.1.4 Universities use data science in their research and also to enhance the study experience of their students. The massive open online courses (MOOC) produces a lot of data, which allows universities to study how this type of learning can complement traditional classes. Examples of MOOCs are Coursera, Udacity, and edX.

3. Facets of data

In data science and big data, there are many different types of data. Each of them tends to require different tools and techniques.

The main categories of data are these:

- Structured
- Unstructured
- Natural language
- Machine-generated
- Graph-based
- Audio, video, and images
- Streaming

3.1 Structured data

Structured data is data that depends on a data model and resides in a fixed field within a record.

It is easy to store structured data in tables within databases or Excel files as shown in figure 1.1. SQL or Structured Query Language, is the preferred way to manage and query data that resides in databases.

There can be structured data that is difficult to store in a traditional relational database. Hierarchical data such as a family tree is one such example. More often, data comes unstructured.

1	Indicator ID	Dimension List	Timeframe	Numeric Value	Missing Value Flag	Confidence Int
2	214390830	Total (Age-adjusted)	2008	74.6%		73.8%
3	214390833	Aged 18-44 years	2008	59.4%		58.0%
4	214390831	Aged 18-24 years	2008	37.4%		34.6%
5	214390832	Aged 25-44 years	2008	66.9%		65.5%
6	214390836	Aged 45-64 years	2008	88.6%		87.7%
7	214390834	Aged 45-54 years	2008	86.3%		85.1%
8	214390835	Aged 55-64 years	2008	91.5%		90.4%
9	214390840	Aged 65 years and over	2008	94.6%		93.8%
10	214390837	Aged 65-74 years	2008	93.6%		92.4%
11	214390838	Aged 75-84 years	2008	95.6%		94.4%
12	214390839	Aged 85 years and over	2008	96.0%		94.0%
13	214390841	Male (Age-adjusted)	2008	72.2%		71.1%
14	214390842	Female (Age-adjusted)	2008	76.8%		75.9%
15	214390843	White only (Age-adjusted)	2008	73.8%		72.9%
16	214390844	Black or African American only (Age-adjusted)	2008	77.0%		75.0%
17	214390845	American Indian or Alaska Native only (Age-adjusted)	2008	66.5%		57.1%
18	214390846	Asian only (Age-adjusted)	2008	80.5%		77.7%
19	214390847	Native Hawaiian or Other Pacific Islander only (Age-adjusted)	2008	DSU		
20	214390848	2 or more races (Age-adjusted)	2008	75.6%		69.6%

Figure 1.1 An Excel table is an example of structured data.

3.2 Unstructured data

Unstructured data is data that isn't easy to fit into a data model because the content is context-specific or varying. One example of unstructured data is the regular email as shown in figure 1.2. Although email contains structured elements such as the sender, title and body text, it is a challenge to find the number of people who have written an email complaint about a specific employee because so many ways exist to refer to a person. For example, The thousands of different languages and dialects out there further complicate this.

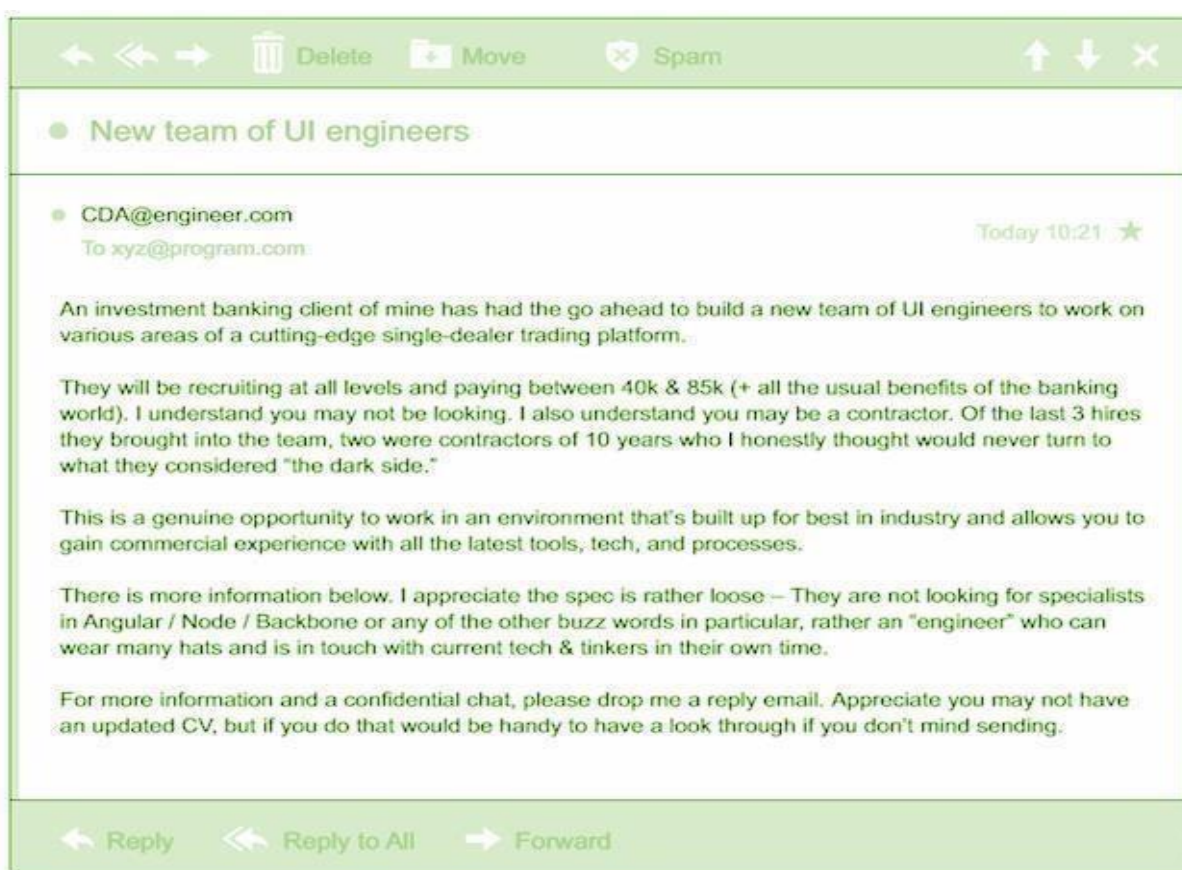


Figure 1.2 Email is simultaneously an example of unstructured data and natural language data.

3.3 Natural language

Natural language is a special type of unstructured data. It is challenging to process because it requires knowledge of specific data science techniques and linguistics. A human-written email as shown in figure 1.2 is also an example of natural language data.

The natural language processing community has had success in entity recognition, topic recognition, summarization, text completion, and sentiment analysis, but models trained in one domain don't generalize well to other domains. Even state-of-the-art techniques aren't able to decipher the meaning of every piece of text as it is ambiguous by nature.

3.4 Machine-generated data

Machine-generated data is information that is automatically created by a computer, process, application, or other machine without human intervention. Machine generated data is becoming a major data resource and will continue to do so. Wiki bon has forecast that the market value of the industrial Internet (a term coined by Frost & Sullivan to refer to the integration of complex physical machinery with networked sensors and software) will be approximately \$540 billion in 2020. IDC (International Data Corporation) has estimated there will be 26 times more connected things than people in 2020. This network is commonly referred to as the internet of things.

The analysis of machine data relies on highly scalable tools, due to its high volume and speed. Examples of machine data are web server logs, call detail records, network event logs and telemetry as shown in figure 1.3.

The machine data shown in figure 1.3 would fit in a classic table-structured database. This isn't the best approach for highly interconnected or "networked" data, where the relationships between entities have a valuable role to play.

CSIPERF:TXCOMMIT;313236	
2014-11-28 11:36:13, Info	CSI 00000153 Creating NT transaction (seq
69), objectname [6]"(null)"	
2014-11-28 11:36:13, Info	CSI 00000154 Created NT transaction (seq 69)
result 0x00000000, handle @0x4e54	
2014-11-28 11:36:13, Info	CSI 00000155@2014/11/28:10:36:13.471
Beginning NT transaction commit...	
2014-11-28 11:36:13, Info	CSI 00000156@2014/11/28:10:36:13.705 CSI perf
trace:	
CSIPERF:TXCOMMIT;273983	
2014-11-28 11:36:13, Info	CSI 00000157 Creating NT transaction (seq
70), objectname [6]"(null)"	
2014-11-28 11:36:13, Info	CSI 00000158 Created NT transaction (seq 70)
result 0x00000000, handle @0x4e5c	
2014-11-28 11:36:13, Info	CSI 00000159@2014/11/28:10:36:13.764
Beginning NT transaction commit...	
2014-11-28 11:36:14, Info	CSI 0000015a@2014/11/28:10:36:14.094 CSI perf
trace:	
CSIPERF:TXCOMMIT;386259	
2014-11-28 11:36:14, Info	CSI 0000015b Creating NT transaction (seq
71), objectname [6]"(null)"	
2014-11-28 11:36:14, Info	CSI 0000015c Created NT transaction (seq 71)
result 0x00000000, handle @0x4e5c	
2014-11-28 11:36:14, Info	CSI 0000015d@2014/11/28:10:36:14.106
Beginning NT transaction commit...	
2014-11-28 11:36:14, Info	CSI 0000015e@2014/11/28:10:36:14.428 CSI perf
trace:	
CSIPERF:TXCOMMIT;375581	

Figure 1.3 Example of machine-generated data

3.5 Graph-based or network data

In graph theory, a graph is a mathematical structure to model pairwise relationships between objects. Graph or network data is the data that focuses on the relationship or adjacency of objects. The graph structures use nodes, edges, and properties to represent and store graphical data. Graph-based data is a natural way to represent social networks, and its structure allows to calculate specific metrics such as the influence of a person and the shortest path between two people. Examples of graph-based data can be found on many social media websites as shown in the figure 1.4.

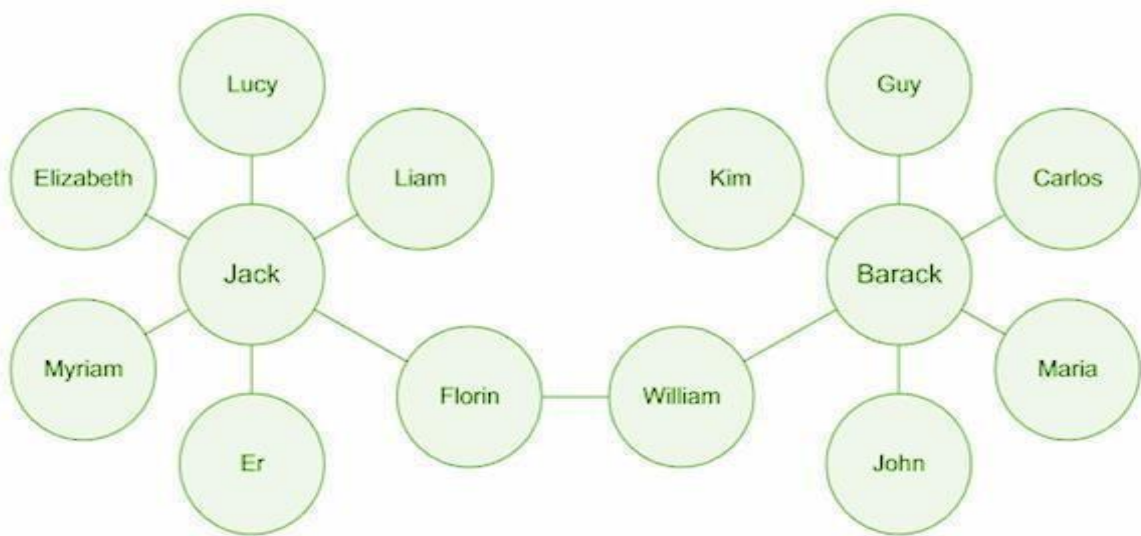


Figure 1.4 Friends in a social network are an example of graph-based data.

Graph databases are used to store graph-based data and are queried with specialized query languages such as SPARQL. Graph data poses its challenges, but for a computer interpreting additive and image data, it can be even more difficult.

3.6 Audio, image and video

Audio, image and video are data types that pose specific challenges to a data scientist. Tasks that are trivial for humans, such as recognizing objects in pictures, turn out to be challenging for computers. In game analytics, High-speed cameras at stadiums will capture ball and athlete movements to calculate in real time, for example, the path taken by a defender relative to two baselines.

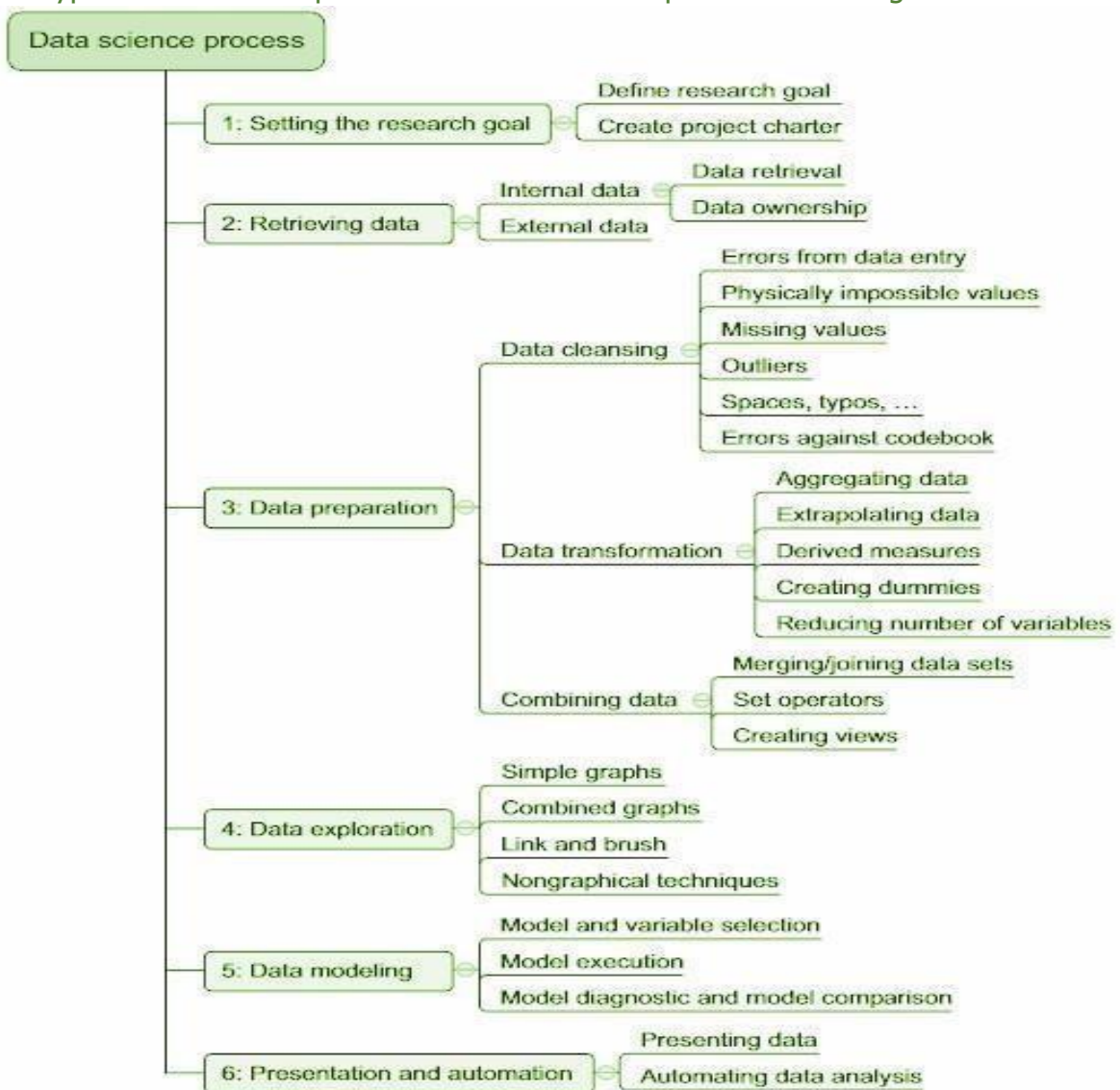
A company called DeepMind succeeded at creating an algorithm that's capable of learning how to play video games. This algorithm takes the video screen as input and learns to interpret everything via a complex process of deep learning which prompted Google to buy the company for their own Artificial Intelligence (AI) development plans. The learning algorithm takes in data as it's produced by the computer game; it's streaming data.

3.7 Streaming data

While streaming data can take almost any of the previous forms, it has an extra property. The data flows into the system when an event happens instead of being loaded into a data store in a batch. Although this isn't really a different type of data, we treat it here as such because we need to adapt our process to deal with this type of information. Examples are the "What's trending" on Twitter, live sporting or music events, and the stock market.

4. Overview of data science process

The typical data science process consists of six steps as shown in figure 4.1.



The data science process and shows the main steps and actions.

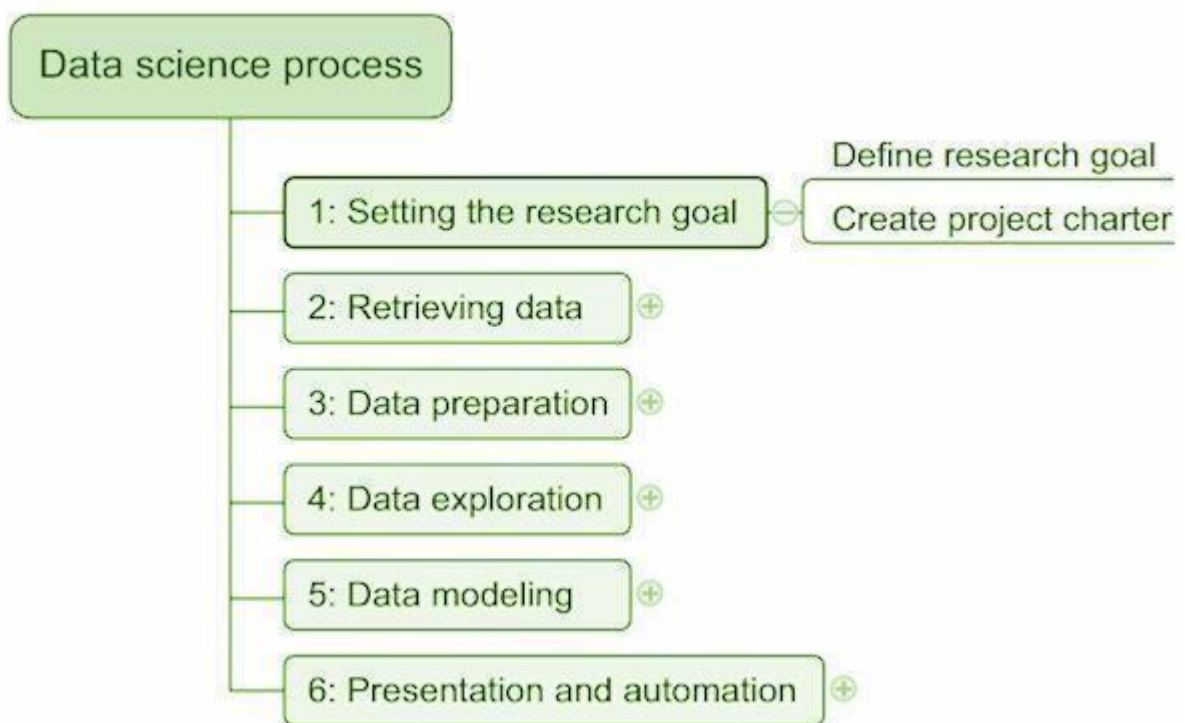
1. The first step of this process is setting a research goal. The main purpose here is making sure all the stakeholders understand the what, how, and why of the project. In every serious project this will result in a project charter.
2. The second phase is data retrieval. The data is required for analysis, so this step includes finding suitable data and getting access to the data from the data owner. The result is data in its raw form, which probably needs transformation before it becomes usable.
3. The third step is data preparation. This includes transforming the data from a raw form into data that is directly usable in the models. To achieve this, detect and correct different kinds of errors in the data, combine data from different data sources and transform it. After completing this step, we can progress to data visualization and modeling.
4. The fourth step is data exploration. The goal of this step is to gain a deep understanding of the data, look for patterns, correlations, deviations based on visual and descriptive techniques. The insights gained from this phase will enable us to start modeling.
5. The fifth step is data modeling. It is now that you attempt to gain the insights or make the predictions stated in your project charter.
6. The last step of the data science model is presenting the results and automating the analysis, if needed. One goal of a project is to change a process and/or make better decisions. The importance of this step is more apparent in projects on a strategic and tactical level. Certain projects require performing the business process over and over again, so automating the project will save time.

In reality we won't progress in a linear way from step 1 to step 6. Often we'll regress and iterate between the different phases. Following these six steps will lead to higher project success ratio and increased impact of research results. This process ensures that we have a well-defined research plan, a good understanding of the business

question and clear deliverables before even starting to look at the data. The first steps of the process focus on getting high-quality data as input for the models. This way the models will perform better later on.

Step 1: Defining research goals and creating a project charter

A project starts by understanding the what, the why, and the how of our project as shown in the figure 2.2. What does the company expect you to do? And why does management place such a value on your research? Answering these three questions (what,why,how) is the goal of the first phase.



The outcome should be a clear research goal, a good understanding of the context, well-defined deliverables and a plan of action with a timetable. This information is then best placed in a project charter. The length and formality differ between projects and companies.

1.1 Spend time understanding the goals and context of your research

An essential outcome is the research goal that states the purpose of your assignment in a clear and focused manner. Understanding the business goals and context is critical for project success.

1.2 Create a project charter

Clients like to know upfront what they are paying for, so after getting a good understanding of the business problem, try to get a formal agreement on the deliverables. All this information is best collected in a project charter. For any significant project this would be mandatory.

A project charter requires teamwork and the inputs should cover the following:

- A clear research goal
- The project mission and context
- How to perform analysis
- What resources are expected to use
- Proof that it's an achievable project or proof of concepts
- Deliverables and a measure of success
- A timeline that the client can use this information to make an estimation of the project costs and the data and people required for the project to become a success.

Step 2: Retrieving data

The next step in data science is to retrieve the required data as shown in figure 2.3. Sometimes there is a need to go into the field and design a data collection process. Many companies will have already collected and stored the data and what they do with it.

have can often be bought from third parties. The organizations are also making high-quality data freely available for public and commercial use.

Data can be stored in many forms, ranging from simple text files to tables in a database. The objective is acquiring all the data needed. This may be difficult, and even if you succeed, data needs polishing to be of any use to you.

1.1 Start with data stored within the company (Internal data)

The first step is to assess the relevance and quality of the data that is readily available within the company. Most companies have a program for maintaining key data, so much of the cleaning work may already be done. This data can be stored in official data repositories such as databases, data marts, data warehouses, and data lakes maintained by a team of IT professionals. The primary goal of a database is data storage, while a data warehouse is designed for reading and analyzing that data.



A data mart is a subset of the data warehouse and geared toward serving a specific business unit. While data warehouses and data marts are home to preprocessed data, data lakes contain data in its natural or raw format. Getting access to data is another difficult task. Organizations understand the value and sensitivity of data and often have policies in place so everyone has access to what they need and nothing more. These policies translate into physical and digital barriers called Chinese walls. These

“walls” are mandatory and well-regulated for customer data in most countries. Getting access to the data may take time and involve company politics.

1.2 Don't be afraid to shop around (External data)

If data isn't available inside your organization, look outside your organization's walls. Many companies specialize in collecting valuable information. Other companies provide data so that you, in turn, can enrich their services and ecosystem. Such is the case with Twitter, LinkedIn, and Facebook. Although data is considered an asset more valuable by certain companies, more and more governments and organizations share their data for free with the world. This data can be of excellent quality and it depends on the institution that creates and manages it. The information they share covers a broad range of topics such as the number of accidents or amount of drug abuse in a certain region and its demographics. This data is helpful when you want to enrich proprietary data but also convenient when training your data science skills. Table 2.1 shows only a small selection from the growing number of open-data providers.

Open data site	Description
Data.gov	The home of the US Government's open data
https://open-data.europa.eu/	The home of the European Commission's open data
Freebase.org	An open database that retrieves its information from sites like Wikipedia, MusicBrains, and the SEC archive
Data.worldbank.org	Open data initiative from the World Bank
Aiddata.org	Open data for international development
Open.fda.gov	Open data from the US Food and Drug Administration

1.3 Do data quality checks to prevent problems

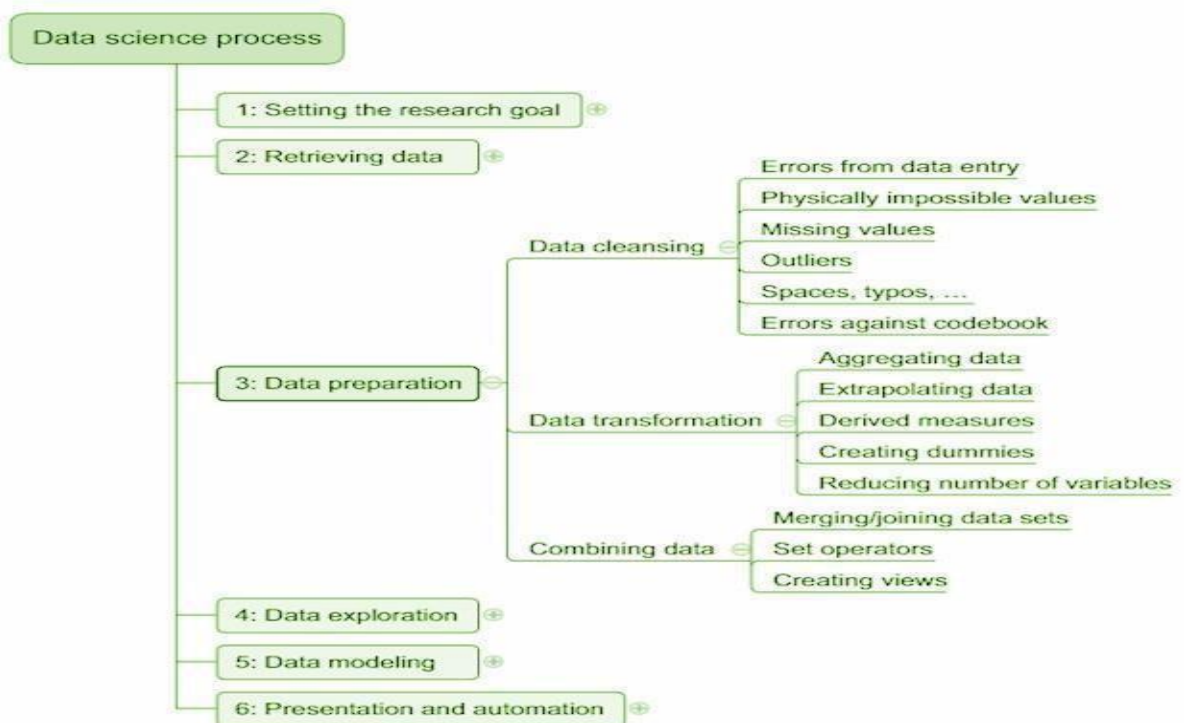
A good portion of the project time should be spent on doing data correction and cleansing. The retrieval of data is the first time we'll inspect the data in the data science process. Most of the errors encountered during the data gathering phase are easy to spot. The data is investigated during the import, data preparation and exploratory phases. The difference is in the goal and the depth of the investigation. During data

retrieval, check to see if the data is equal to the data in the source document and to see if we have the right data types.

Step 3: Cleansing, integrating and transforming data (Data preparation)

The data received from the data retrieval phase is likely to be “a diamond in the rough.” so the task is to prepare it for use in the modeling and reporting phase.

The model needs the data in specific format, so data transformation will always come into play. It is good to correct data errors as early on in the process as possible. Figure 1.4 shows the most common actions to take during the data cleansing,



integration, and transformation phase.

3.1 Cleansing data

Data cleansing is a subprocess of the data science process that focuses on removing errors in the data so that the data becomes a true and consistent representation of the processes it originates from. There are at least two types of errors. The first type is the interpretation error, such as when you take the value in your data for granted, like saying that a person’s age is greater than 300 years. The second type of error is

points to inconsistencies between data sources or against your company's standardized values. The table below shows an overview of the types of errors that can be detected with easy checks.

Errors pointing to false values within one dataset

Error description	Possible solution
Mistakes during data entry	Manual overrules
Redundant white space	Use string functions
Impossible values	Manual overrules
Missing values	Remove observation or value
Outliers	Validate and, if erroneous, treat as missing value (remove or insert)

Errors pointing to inconsistencies between data sets

Error description	Possible solution
Deviations from a code book	Match on keys or else use manual overrules
Different units of measurement	Recalculate
Different levels of aggregation	Bring to same level of measurement by aggregation or extrapolation

General solution: Try to fix the problem early in the data acquisition chain or else fix it in the program. Sometimes there is a need to use more advanced methods, such as simple modeling, to find and identify data errors; diagnostic plots can be especially insightful.

3.1.1 Data entry errors

Data collection and data entry are error-prone processes. They often require human intervention, as humans make typos or lose their concentration for a second and introduce an error into the chain. But data collected by machines or computers isn't free from errors either. Errors can arise from human sloppiness, whereas others are due to machine or hardware failure. Examples of errors originating from machines are transmission errors or bugs in the extract, transform, and load phase (ETL).

For small data sets we can check every value by hand. Detecting data errors when the variables we study don't have many classes can be done by tabulating the data with counts. When we have a variable that can take only two values: "Good" and "Bad", we can create a frequency table and see if those are truly the only two values present.

In table 2.3, the values "Godo" and "Bade" point out something went wrong in at least 16 cases.

Value	Count
Good	1598647
Bad	1354468
Godo	15
Bade	1

Table 2.3 Detecting outliers on simple variables with a frequency table

Most errors of this type are easy to fix with simple assignment statements and ifthen else rules:

```
if x == "Godo": x =  
    "Good"  
if x == "Bade":  
    x = "Bad"
```

3.1.2 Redundant whitespace

Whitespaces tend to be hard to detect but cause errors like other redundant characters would. Fixing redundant whitespaces can be done by most programming languages.

They all provide string functions that will remove the leading and trailing whitespaces.

For instance, in Python you can use the `strip()` function to remove leading and trailing spaces.

Fixing capital letter mismatches

Capital letter mismatches are common. Most programming languages make a distinction between "Brazil" and "brazil". In this case we can solve the problem by applying a function that returns both strings in lowercase, such as `.lower()` in Python. `"Brazil".lower() == "brazil".lower()` should result in `true`.

3.1.3 Impossible values and sanity checks

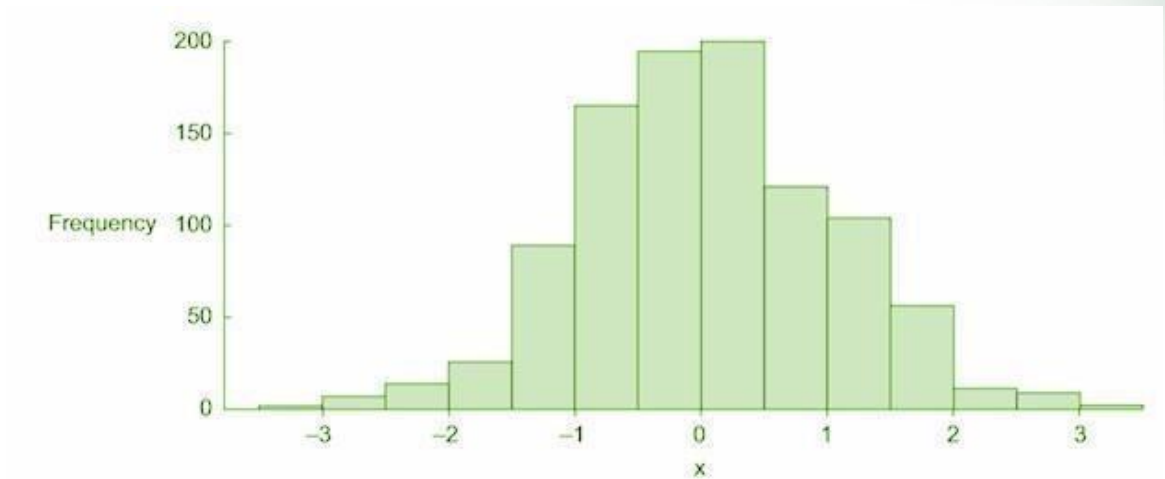
Sanity checks are another valuable type of data check. Here we check the value against physically or theoretically impossible values such as people taller than 3 meters or someone with an age of 299 years. Sanity checks can be directly expressed with rules:

```
check = 0 <= age <= 120
```

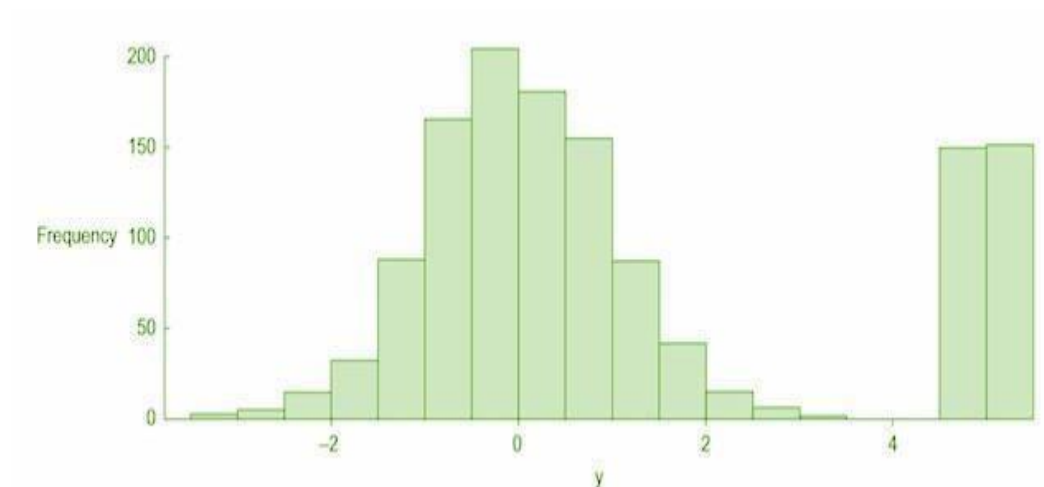
3.1.4 Outliers

An outlier is an observation that seems to be distant from other observations or, more specifically, one observation that follows a different logic or generative process than the other observations. The easiest way to find outliers is to use a plot or a table with the minimum and maximum values. An example is shown in figure 2.6. The plot on the top shows no outliers, whereas the plot on the bottom shows possible outliers on the upper side when a normal distribution is expected. The normal distribution or Gaussian distribution is the most common distribution.

Expected distribution:



outliers:



- Distribution plots are helpful in detecting outliers and helps in understanding the variable.
- It shows most cases occurring around the average of the distribution and the occurrences decrease when further away from it.
- The high values in the bottom graph can point to outliers when assuming a normal distribution.

3.1.5 Dealing with missing values

Missing values aren't necessarily wrong, but we still need to handle them separately. Certain modeling techniques can't handle missing values. They might be an indicator that something went wrong in data collection or that an error happened in the data.

process. Common techniques used are listed in table. Which technique to use at what time is dependent on a particular case.

Technique	Advantage	Disadvantage
Omit the values	Easy to perform	You lose the information from an observation
Set value to null	Easy to perform	Not every modeling technique and/or implementation can handle null values
Impute a static value such as 0 or the mean	Easy to perform You don't lose information from the other variables in the observation	Can lead to false estimations from a model
Impute a value from an estimated or theoretical distribution	Does not disturb the model as much	Harder to execute You make data assumptions
Modeling the value (nondependent)	Does not disturb the model too much	Can lead to too much confidence in the model Can artificially raise dependence among the variables Harder to execute You make data assumptions

3.1.6 Deviations from a code book

Detecting errors in larger data sets against a code book or against standardized values can be done with the help of set operations. A code book is a description of data, a form of metadata.

It contains things such as the number of variables per observation, the number of observations, and what each encoding within a variable means. For instance “0” equals “negative”, “5” stands for “very positive”. A code book also tells the type of data we are looking at: is it hierarchical, graph, something else.

3.2 Combining data from different data sources

The data comes from several different places and we need to integrate these different sources. Data varies in size, type and structure, ranging from databases and Excel files to text documents.

The different ways of combining the data

There are two operations to combine information from different data sets.

- The first operation is joining: enriching an observation from one table with information from another table.
- The second operation is appending or stacking: adding the observations of one table to those of another table.

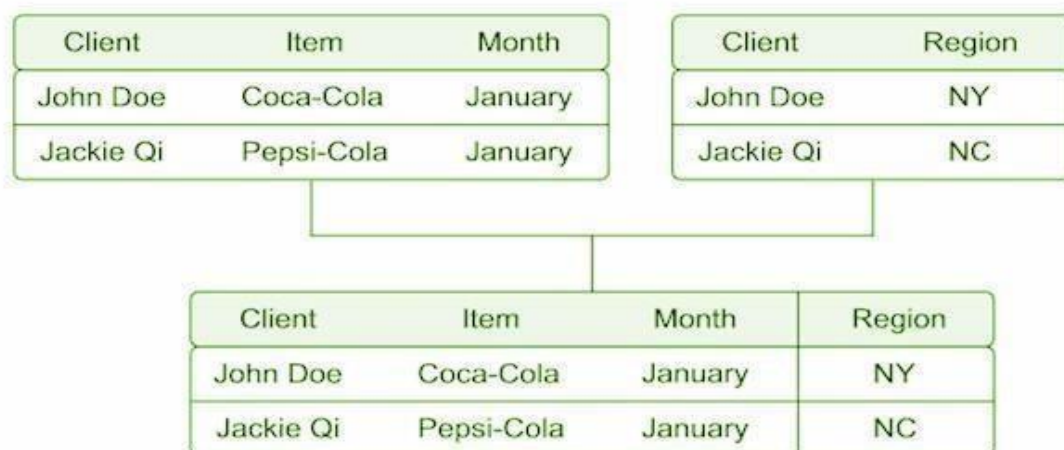
When the data is combined, we have the option to create a new physical table or a virtual table by creating a view. The advantage of a view is that it doesn't consume more disk space.

3.2.1 Joining tables

Joining tables allows us to combine the information of one observation found in one table with the information that is found in another table. The focus is on enriching a single observation.

Let's say that the first table contains information about the purchases of a customer and the other table contains information about the region where your customer lives. Joining the tables allows us to combine the information so that we can use it for a model, as shown in figure 2.7. To join tables, we use variables that represent the same

object in both tables, such as a date, a country name, or a Social Security number. These common fields are known as keys. When these keys also uniquely define



the records in the table they are called primary keys.

Figure 2.7 Joining two tables on the Item and Region keys

One table may have buying behavior and the other table may have demographic information on a person. In figure 2.7 both tables contain the client name, and this makes it easy to enrich the client expenditures with the region of the client.

3.2.2 Appending tables

Appending or stacking tables is effectively adding observations from one table to another table. Figure 2.8 shows an example of appending tables. One table contains the observations from the month January and the second table contains observations from the month February. The result of appending these tables is a larger one with the observations from January as well as February.

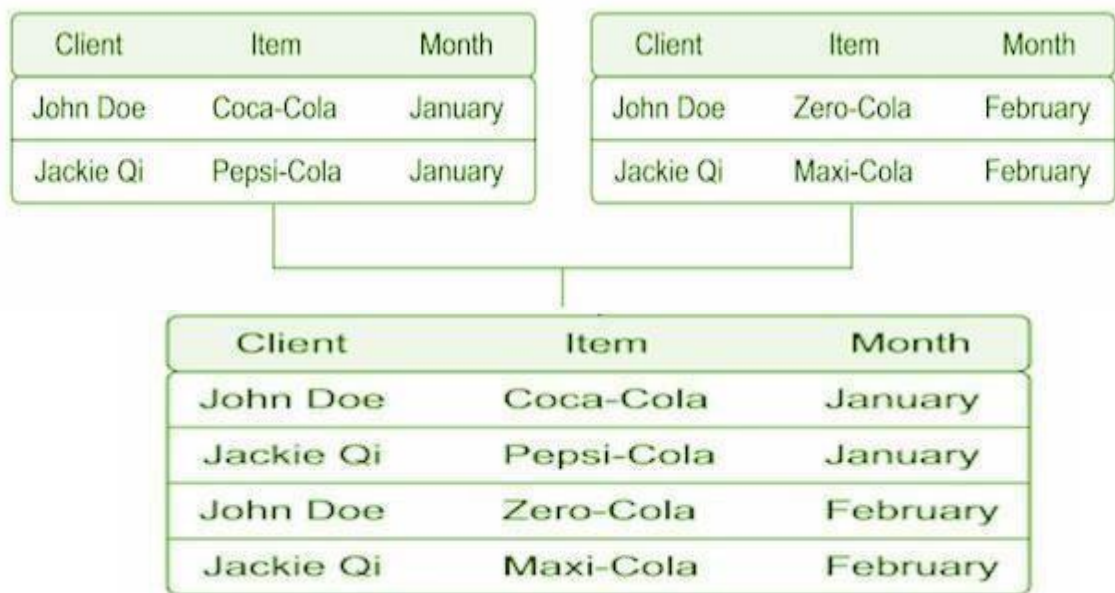


Figure 2.8 Appending data from tables

Appending data from tables is a common operation but requires an equal structure in the tables being appended.

3.2.3 Using views to simulate data joins and appends

To avoid duplication of data, we virtually combine data with views. The problem is that when we duplicate the data, more storage space is needed. In case of less data, that may not cause problems. But if every table consists of terabytes of data, then it becomes problematic to duplicate the data. For this reason, the concept of a view was invented. A view behaves as if we are working on a table, but this table is nothing but a virtual layer that combines the tables. Figure 2.9 shows how the sales data from the different months is combined virtually into a yearly sales table instead of duplicating the data. Views do come with a drawback, however. While a table join is only performed once, the join that creates the view is recreated every time it's queried, using more processing power than a pre-calculated table would have.



Figure 2.9 A view helps to combine data without replication.

3.2.4 Enriching aggregated measures

Data enrichment can also be done by adding calculated information to the table, such as the total number of sales or what percentage of total stock has been sold in a certain region as shown in the figure below. We now have an aggregated data set, which in turn can be used to calculate the participation of each product within its category.

Product class	Product	Sales in \$	Sales t-1 in \$	Growth	Sales by product class	Rank sales
A	B	X	Y	$(X-Y) / Y$	AX	NX
Sport	Sport 1	95	98	-3.06%	215	2
Sport	Sport 2	120	132	-9.09%	215	1
Shoes	Shoes 1	10	6	66.67%	10	3

3.3 Transforming data

Certain models require their data to be in a certain shape. After cleansing and integrating the data, the next task is to transform the data so it takes a suitable form for data modeling.

Relationships between an input variable and an output variable aren't always linear. Take, for instance, a relationship of the form $y = aebx$. Taking the log of the independent variables simplifies the estimation problem dramatically. Figure 2.11 shows how transforming the input variables greatly simplifies the estimation problem.

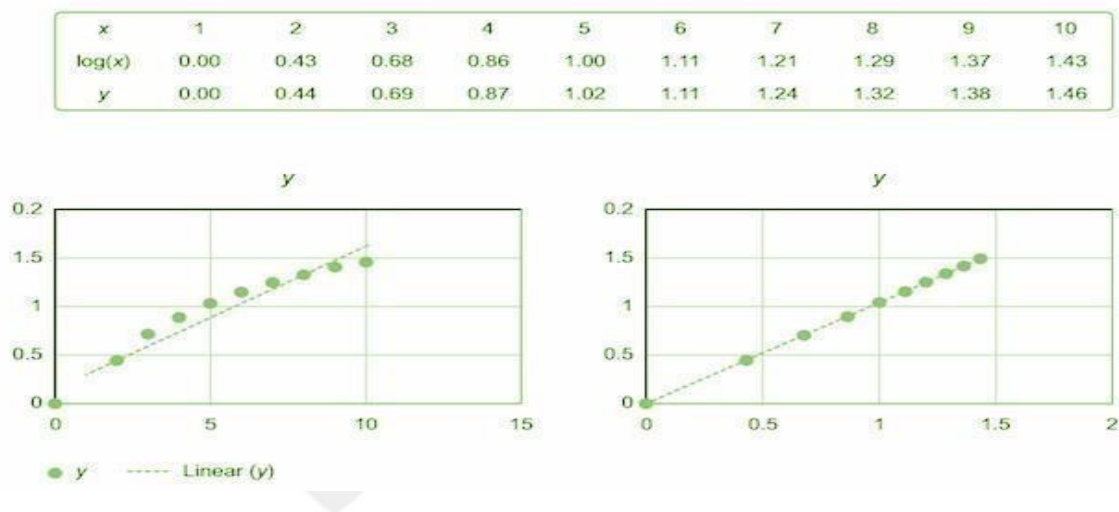


Figure 2.11 Transforming x to $\log x$ makes the relationship between x and y linear (right), compared with the non- $\log x$ (left).

3.3.1 Reducing the number of variables

Sometimes there are too many variables and we need to reduce the number because they don't add new information to the model. Having too many variables in the model makes the model difficult to handle, and certain techniques don't perform well when we overload them with too many input variables. For instance, all the techniques based on a Euclidean distance perform well only up to 10 variables.

3.3.2 Turning variables into dummies

Variables can be turned into dummy variables as shown in figure 2.13. Dummy variables can only take two values: true(1) or false(0). They are used to indicate the absence of a categorical effect that may explain the observation. In this case there are separate columns for the classes stored in one variable and indicate it with 1 if the class is present and 0 otherwise. An example is turning one column named Weekdays into the columns Monday through Sunday. we use an indicator to show if the observation was on a Monday; put 1 on Monday and 0 elsewhere. Turning variables



into

dummies is a technique that is used in modeling.

Figure 2.13 Turning variables into dummies

Turning variables into dummies is a data transformation that breaks a variable that has multiple classes into multiple variables, each having only two possible values: 0 or 1.

Step 4: Exploratory data analysis

Information becomes much easier to grasp when shown in a picture, therefore you mainly use graphical techniques to gain an understanding of your data and the interactions between variables. This phase is about exploring data, discovering anomalies missed before and to fix them. The visualization techniques used in this phase range from simple line graphs or histograms as shown in figure 2.15, to more complex diagrams such as Sankey and network graphs. Sometimes it's useful to compose a composite graph from simple graphs to get even more insight into the data. Other times the graphs can be animated or made interactive to make it easier.

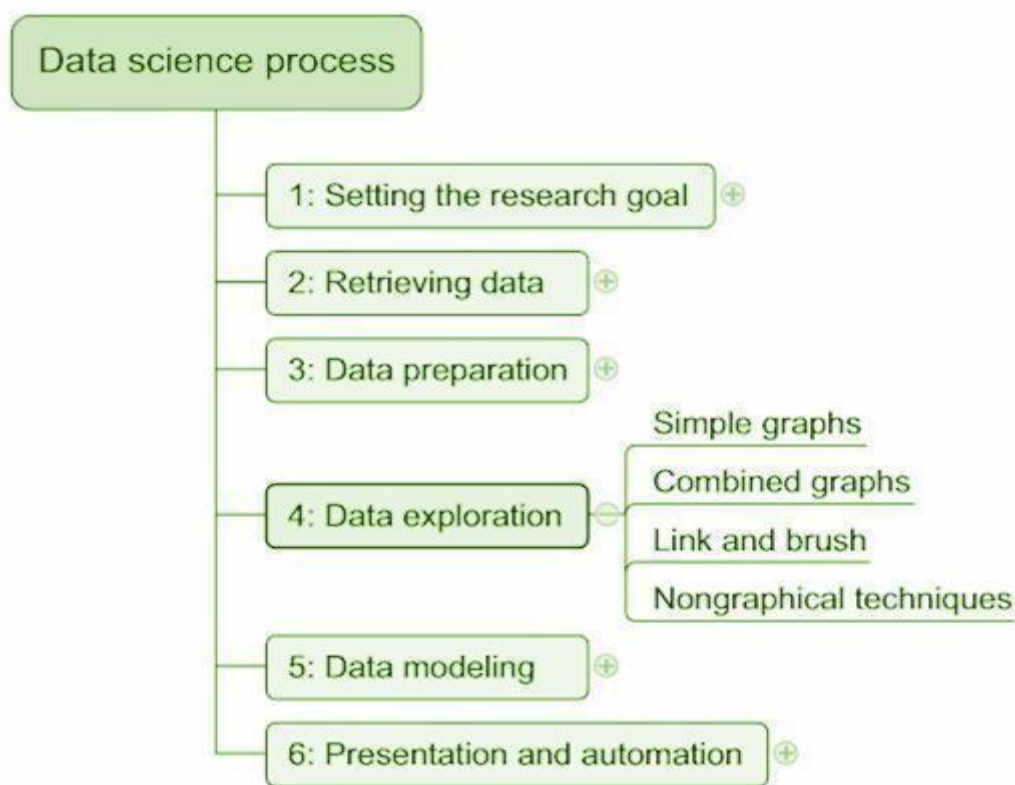


Figure 2.14 Step 4: Data exploration



Figure 2.15 From top to bottom, a bar chart, a line plot, and a distribution are some of the graphs used in exploratory analysis.

These plots can be combined to provide even more insight, as shown in figure 2.16.

Overlaying several plots is common practice.

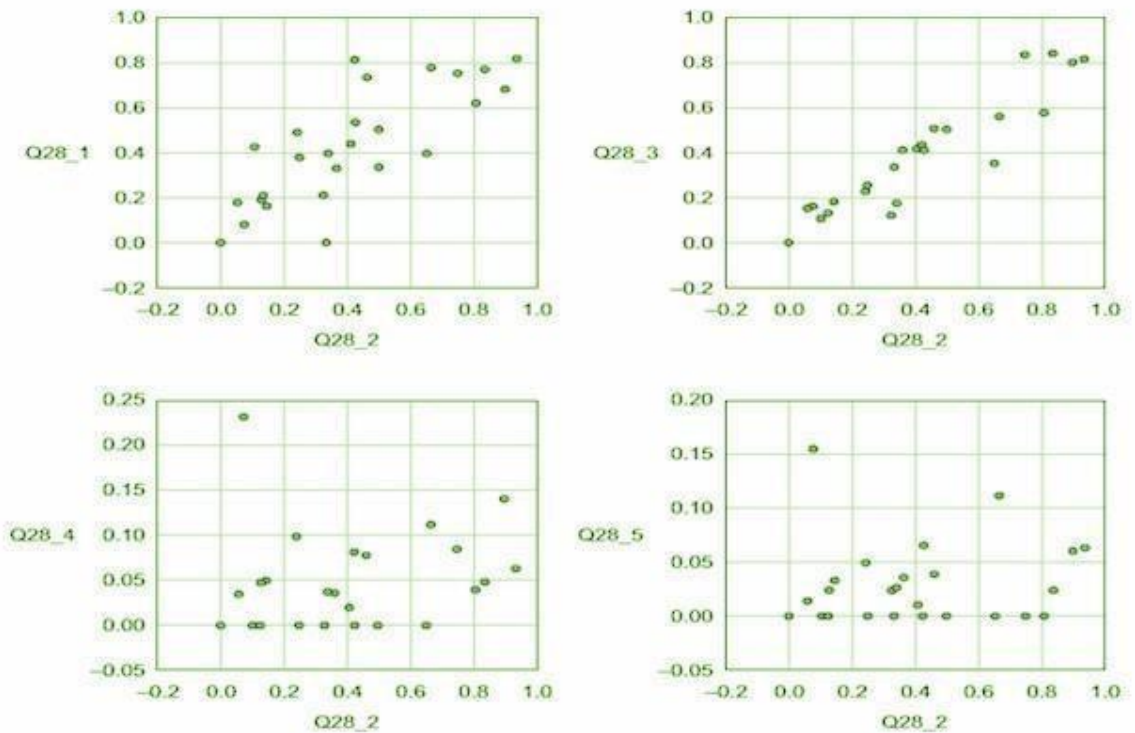
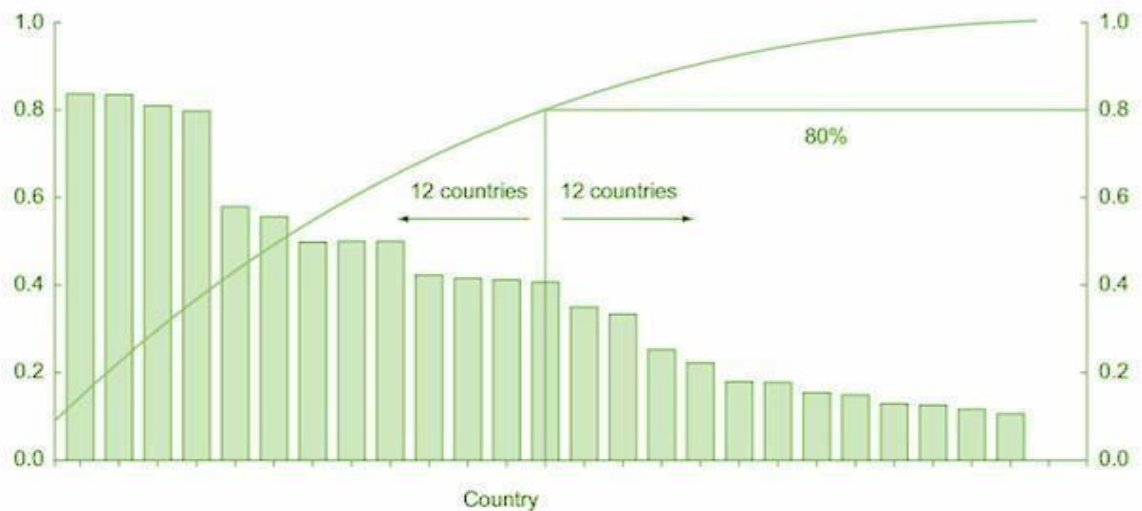
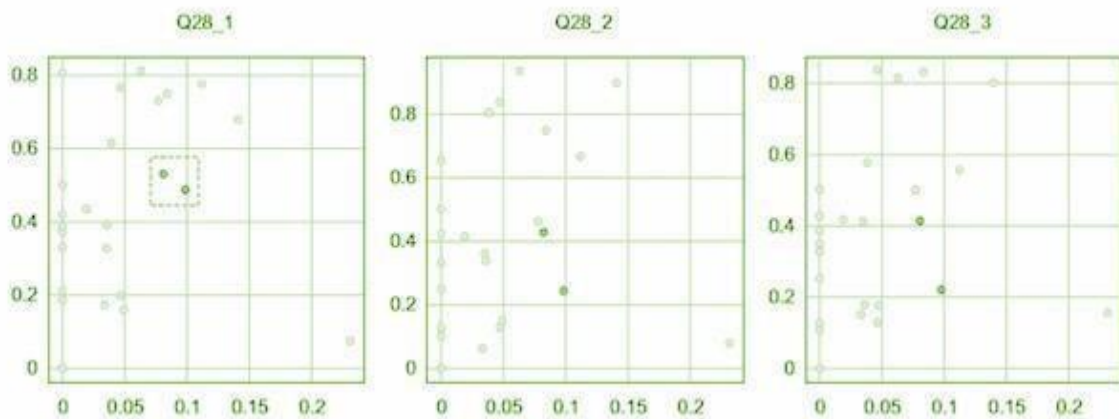


Figure 2.16 Drawing multiple plots together helps to understand the structure of the data over multiple variables.



A Pareto diagram is a combination of the values and a cumulative distribution. It's easy to see from this diagram that the first 50% of the countries contain slightly less than



80% of the total amount.

Figure 2.18 Link and brush allows us to select observations in one plot and highlight the same observations in the other plots.

Figure 2.18 shows another technique: brushing and linking. With brushing and linking we combine and link different graphs and tables (or views) so changes in one graph are automatically transferred to the other graphs. This interactive exploration of data facilitates the discovery of new insights.

Figure 2.18 shows the average score per country for questions. Not only does this indicate a high correlation between the answers, but it's easy to see that when we select several points on a subplot, the points will correspond to similar points on the other graphs. In this case the selected points on the left graph correspond to points on the middle and right graphs, although they correspond better in the middle and right graphs.

Two other important graphs are the histogram shown in figure 2.19 and the boxplot shown in figure 2.20. In a histogram a variable is cut into discrete categories and the number of occurrences in each category are summed up and shown in the graph.

The boxplot, on the other hand, doesn't show how many observations are present but does offer an impression of the distribution within categories. It can

show the maximum, minimum, median, and other characterizing measures at the same time.

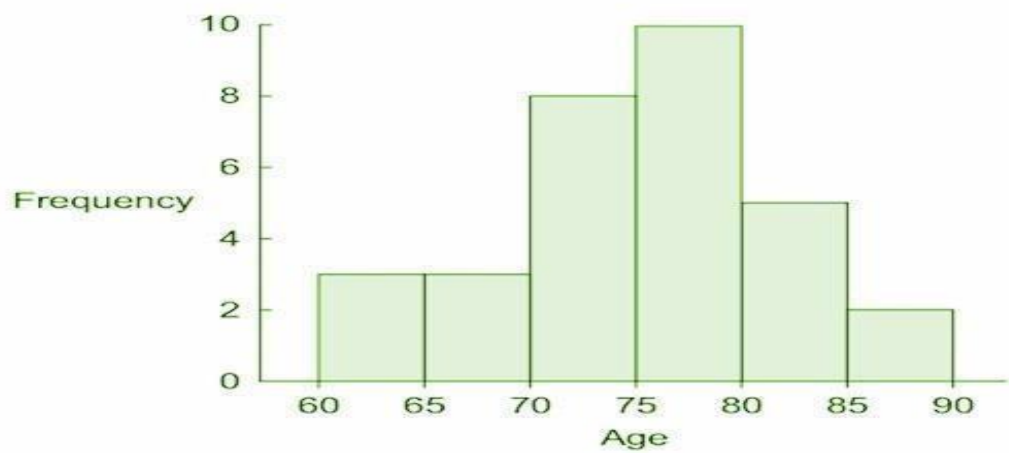
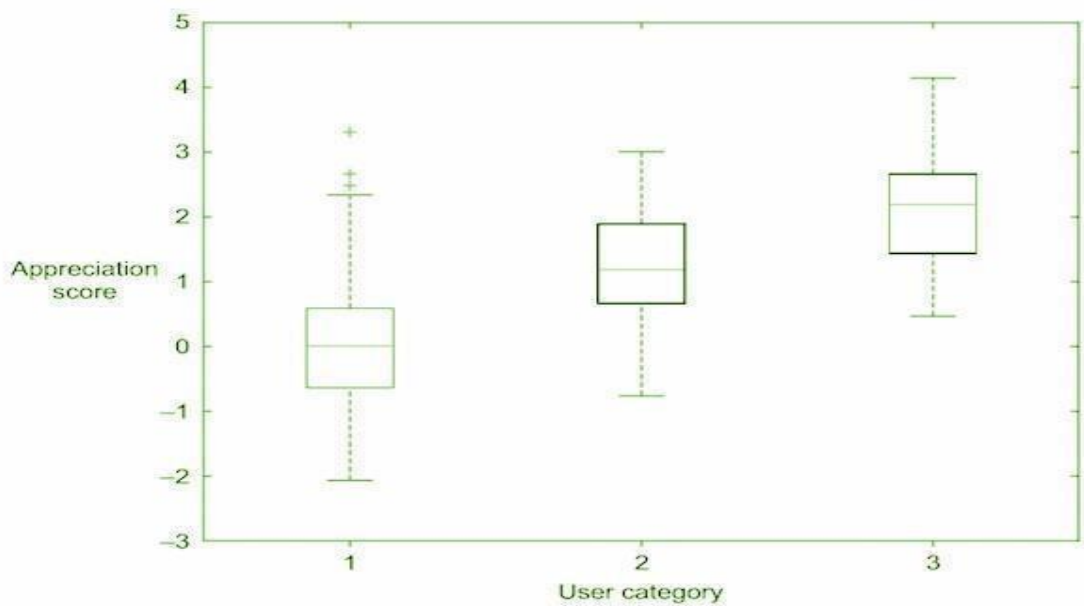


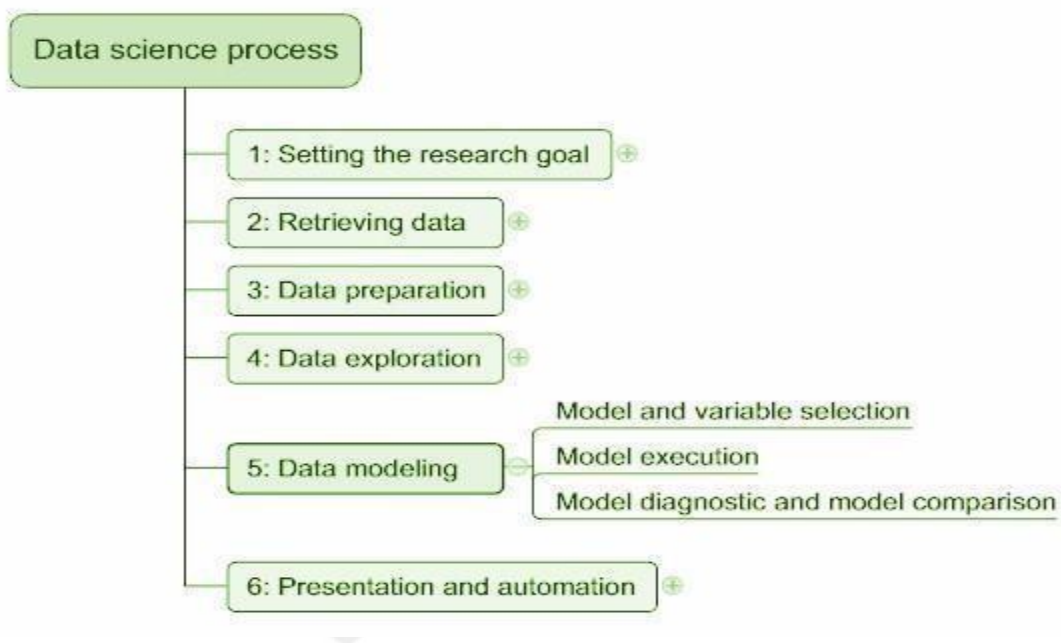
Figure 2.19 Example histogram: the number of people in the age groups of 5-year intervals



These techniques are mainly visual, but in practice they're certainly not limited to visualization techniques. Tabulation, clustering, and other modeling techniques can also be a part of exploratory analysis. Even building simple models can be a part of this step.

Step 5: Build the models

With clean data in place and a good understanding of the content, we are ready to build models with the goal of making better predictions, classifying objects, or gaining an understanding of the system that we are modeling. This phase is much more focused than the exploratory analysis step, because we know what we are looking for and what we want the outcome to be. Figure 2.21 shows the components



of model building.

Figure 2.21 Step 5: Data modeling

Building a model is an iterative process. Most of the models consist of the following main steps:

1. Selection of a modeling technique and variables to enter in the model
2. Execution of the model
3. Diagnosis and model comparison

5.1 Model and variable selection

We need to select the variables we want to include in our model and a modeling technique. Many modeling techniques are available, we need to choose the right model

for a problem. The model performance and all the requirements to use the model has to be considered. The other factors are:

- Must the model be moved to a production environment and, if so, would it be easy to implement?
- How difficult is the maintenance on the model: how long will it remain relevant if left untouched?
- Does the model need to be easy to explain?

5.2 Model execution

Once we have chosen a model we will need to implement it in code. The following listing shows the execution of a linear prediction model.

Listing 2.1 Executing a linear prediction model on semi-random data

```
import statsmodels.api as sm
import numpy as np
predictors = np.random.random(1000).reshape(500,2)
target = predictors.dot(np.array([0.4, 0.6])) + np.random.random(500)
lmRegModel = sm.OLS(target,predictors)
result = lmRegModel.fit()
result.summary()
```

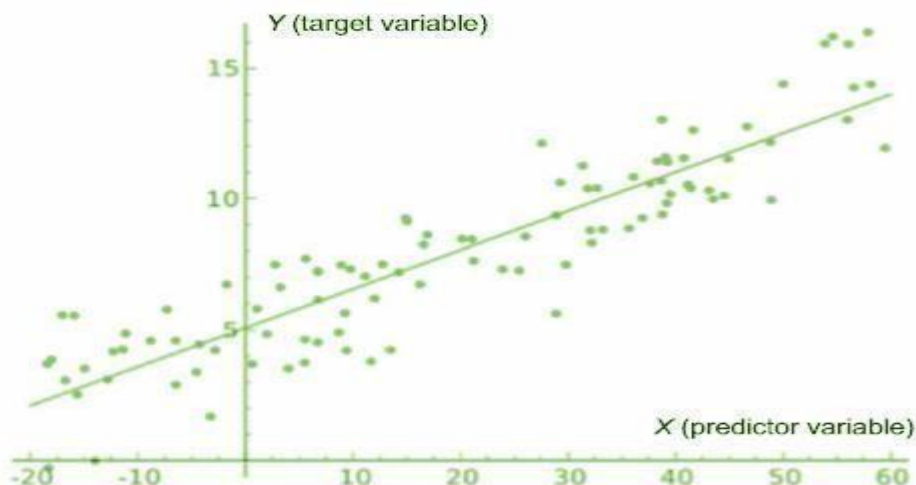
Imports required Python modules.

Fits linear regression on data.

Shows model fit statistics.

Creates random data for predictors (x-values) and semi-random data for the target (y-values) of the model. We use predictors as input to create the target so we infer a correlation here.

Figure 2.22 Linear regression tries to fit a line while minimizing the distance to each



point

We created predictor values that are meant to predict how the target variables behave. For a linear regression, a “linear relation” between each x (predictor) and the y (target) variable is assumed, as shown in figure 2.22. We created the target variable based on the predictor by adding a bit of randomness, this gives us a wellfitting model. The results.summary() outputs the table in figure 2.23. The exact outcome depends on the random variables we got.

Dep. Variable:	y	R-squared:	0.893
Model:	OLS	Adj. R-squared:	0.893
Method:	Least Squares	F-statistic:	2088.
Date:	Fri, 30 Oct 2015	Prob (F-statistic):	7.13e-243
Time:	12:44:31	Log-Likelihood:	-176.74
No. Observations:	500	AIC:	357.5
Df Residuals:	498	BIC:	365.9
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[95.0% Conf. Int.]
x1	0.7658	0.040	19.130	0.000	0.687 0.844
x2	1.1252	0.039	28.603	0.000	1.048 1.202

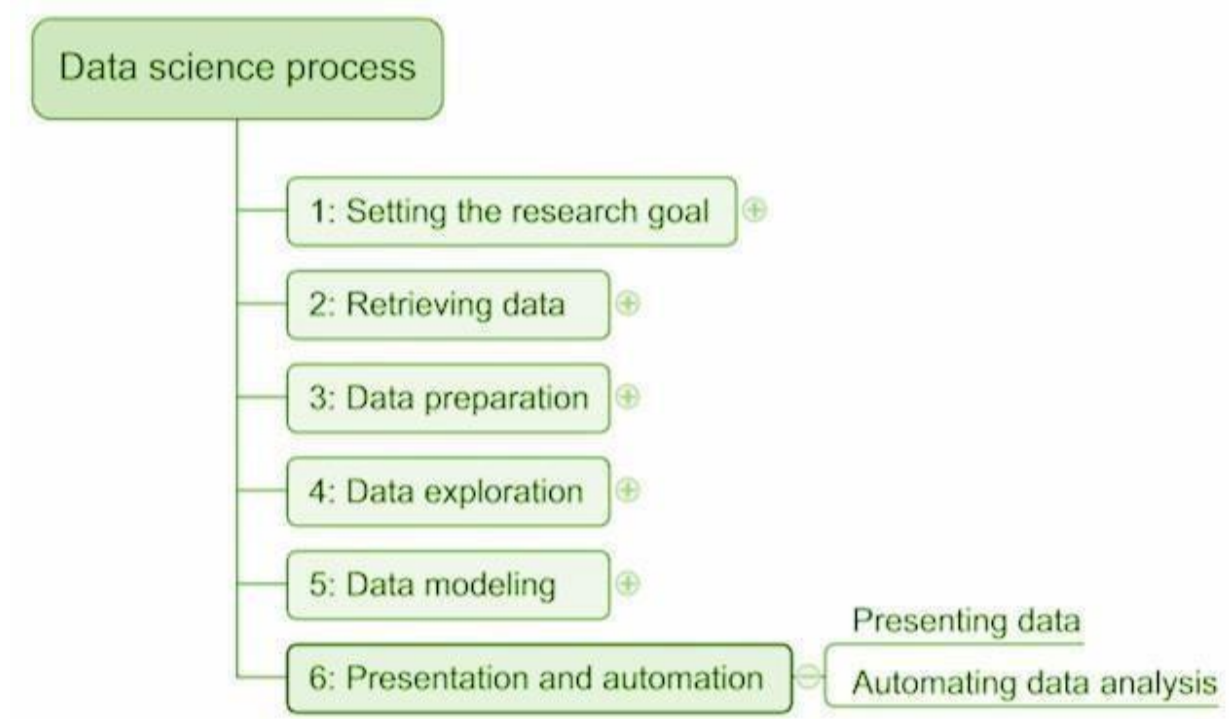
Omnibus:	34.269	Durbin-Watson:	1.943
Prob(Omnibus):	0.000	Jarque-Bera (JB):	13.480
Skew:	-0.125	Prob(JB):	0.00118
Kurtosis:	2.235	Cond. No.	2.51

R-squared and Adj.R-squared is Model fit: higher is better but too high is suspicious

The p-value to show whether a predictor variable has a significant influence on the target. Lower is better and <0.05 is often considered “significant.” Linear equation coefficients. $y = 0.7658x_1 + 1.1252x_2$.

Step 6: Presenting findings and building applications

After analysing the data and building a well-performing model, we have to present the findings as shown in figure 6. The predictions of the models or the insights produced is of great value. For this reason, we need to automate the models. We can also build an application that automatically updates reports, Excel spreadsheets,



or PowerPoint presentations.

Figure 6: Presentation and automation

7. Data Mining

7.1 Why data mining?

We live in a world where vast amounts of data are collected daily. Analyzing such data is an important need. Data mining provides tools to discover knowledge from data. It can be viewed as a result of the natural evolution of information technology. Data mining turns a large collection of data into knowledge.

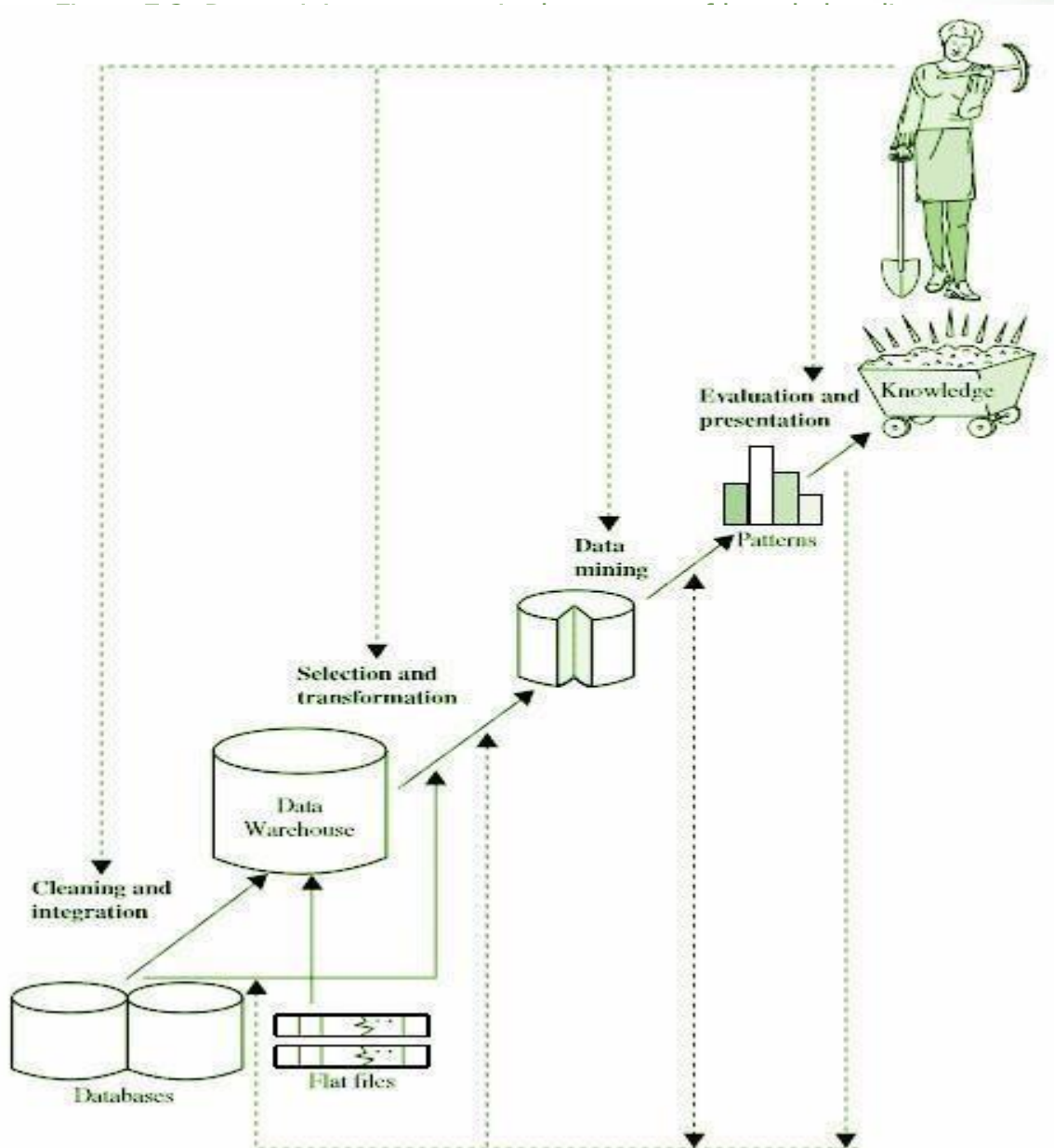
- Terabytes or petabytes of data pour into computer networks, the World Wide Web (WWW), and various data storage devices every day from business, society, science and engineering, medicine, and almost every other aspect of daily life. This growth of available data volume is a result of the computerization of our society and the fast development of powerful data collection and storage tools.
- Businesses worldwide generate gigantic data sets, including sales transactions, stock trading records, product descriptions, sales promotions, company profiles and performance, and customer feedback. For example, large stores, such as Wal-Mart, handle hundreds of millions of transactions per week at thousands of branches around the world.
- Scientific and engineering practices generate high orders of petabytes of data in a continuous manner, from remote sensing, process measuring, scientific experiments, system performance, engineering observations, and environment surveillance. Global backbone telecommunication networks carry tens of petabytes of data traffic everyday.
- The medical and health industry generates tremendous amounts of data from medical records, patient monitoring, and medical imaging.
- Billions of Web searches supported by search engines process tens of petabytes of data daily. Communities and social media have become increasingly important data sources, producing digital pictures and videos, blogs, Web communities and various kinds of social networks.
- The list of sources that generate huge amounts of data is endless. Powerful and versatile tools are needed to automatically uncover valuable

information from the tremendous amounts of data and to transform such data into organized knowledge. This necessity has led to the birth of data mining.

7.2 What is data mining?

Data mining is searching for knowledge (interesting patterns) in data. Data mining is an essential step in the process of knowledge discovery. The knowledge discovery process is shown in Figure 7 as an iterative sequence of the following steps:

1. Data cleaning: To remove noise and inconsistent data
2. Data integration: Where multiple data sources may be combined
3. Data selection: Where data relevant to the analysis task are retrieved from the database
4. Data transformation: Where data are transformed and consolidated into form appropriate for mining by performing summary or aggregation operations.
5. Data mining: An essential process where intelligent methods are applied to extract data patterns.
6. Pattern evaluation: To identify the interesting patterns representing knowledge based on interestingness measures.
7. Knowledge presentation: Where visualization and knowledge representation techniques are used to present mined knowledge to users.



Steps 1 through 4 are different forms of data preprocessing, where data is prepared for mining. The data mining step may interact with the user or a knowledge base. The interesting patterns are presented to the user and may be stored as new knowledge in the knowledge base.

Data mining is the process of discovering interesting patterns and knowledge from large amounts of data. The data sources can include databases, data warehouses,

web, other information repositories or data that are streamed into the system dynamically.

7.3 What kinds of data can be mined ?

The most basic forms of data for mining applications are database data, data warehouse data and transactional data. Data mining can also be applied to other forms of data such as data streams, ordered/sequence data, graph or networked data, spatial data, text data, multimedia data and the WWW.

7.3.1 Database data

A database system, also called a database management system (DBMS), consists of a collection of interrelated data known as a database and a set of software programs to manage and access the data. The software programs provide mechanisms for defining database structures and data storage, for specifying and managing concurrent, shared, or distributed data access and for ensuring consistency and security of the information stored despite system crashes or attempts at unauthorized access.

Relational databases are one of the most commonly available and richest information repositories and thus they are a major data form in the study of data mining.

7.3.2 Data warehouses

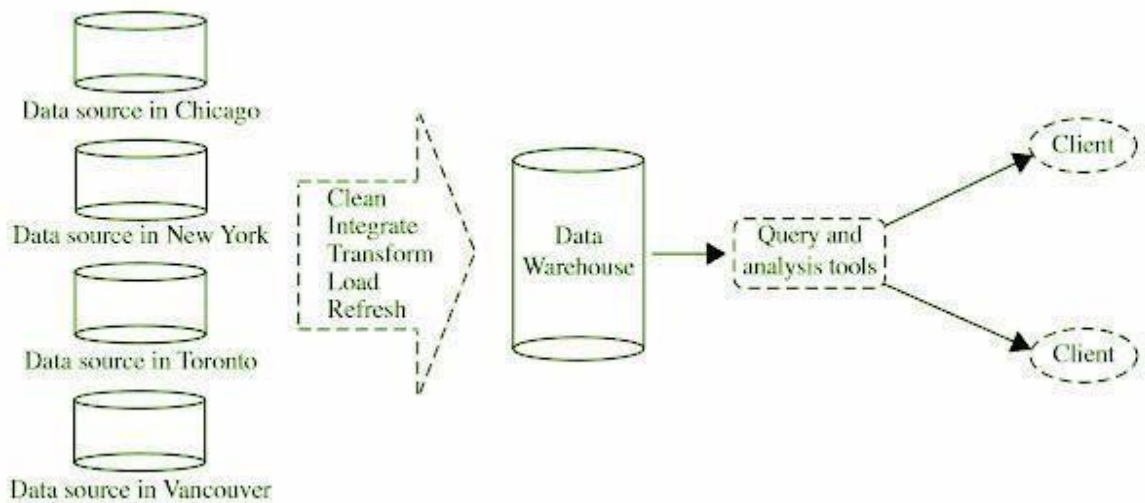
A data warehouse is a repository of information collected from multiple sources stored under a unified schema and usually residing at a single site. Data warehouses are constructed via a process of data cleaning, data integration, data transformation, data loading, and periodic data refreshing.

For example, rather than storing the details of each sales transaction, the data warehouse may store a summary of the transactions per item type for each store or summarized to a higher level for each sales region.

A data warehouse is usually modeled by a multidimensional data structure, called a data cube, in which each dimension corresponds to an attribute or a set of attributes of



in the schema, and each cell stores the value of some aggregate measure such as count or sum(sales amount). A data cube provides a multidimensional view of data and allows the precomputation and fast access of summarized data.



7.3.3 Transactional Data

Each record in a transactional database captures a transaction such as a customer's purchase, a flight booking or a user's clicks on a webpage. A transaction typically includes a unique transaction identity number (trans ID) and a list of the items making up the transaction, such as the items purchased in the transaction. A transactional database may have additional tables, which contain other information related to the transactions, such as item description, information about the salesperson or the branch, and so on.

7.3.4 Other kinds of data

There are many other kinds of data that have versatile forms and structures and rather different semantic meanings. Such kinds of data can be seen in many applications:

- Time-related or sequence data e.g historical records, stock exchange data and time-series and biological sequence data.

- Data streams e.g video surveillance and sensor data, which are continuously transmitted.
- Spatial data e.g maps.
- Engineering design data e.g the design of buildings, system components, or integrated circuits.
- Hypertext and multimedia data including text, image, video, and audio data.
- Graph and networked data e.g social and information networks.
- The Web which is a huge, widely distributed information repository made available by the Internet.

These applications bring about new challenges like how to handle data carrying special structures such as sequences, trees, graphs, and networks and specific semantics such as ordering, image, audio and video contents, and connectivity and how to mine patterns that carry rich structures and semantics.

7.4 What kinds of patterns can be mined?

- Class/Concept Description: Characterization and Discrimination
- Mining Frequent Patterns, Associations, and Correlations
- Classification and Regression for Predictive Analysis
- Cluster Analysis
- Outlier Analysis

7.5 Technologies used in data mining

7.5.1 Statistics

Statistics studies the collection, analysis, interpretation or explanation and presentation of data. Data mining has an inherent connection with statistics. A statistical model is a set of mathematical functions that describe the behavior of the objects in a target class in terms of random variables and their associated probability distributions. Statistical models are widely used to model data and data classes. For example, in data mining tasks like data characterization and classification, statistical models of target classes can be built. Statistical methods can also be used to verify data mining results

7.5.2 Machine Learning

For classification and clustering tasks, machine learning research often focuses on the accuracy of the model. In addition to accuracy, data mining research places strong emphasis on the efficiency and scalability of mining methods on large data sets, as well as on ways to handle complex types of data and explore new, alternative methods.

7.5.3 Database Systems and Data Warehouses

Database systems research focuses on the creation, maintenance, and use of databases for organizations and end-users. Many data mining tasks need to handle large data sets or even real-time, fast streaming data. Therefore, data mining can make good use of scalable database technologies to achieve high efficiency and scalability on large datasets. Data mining tasks can be used to extend the capability of existing database systems to satisfy advanced user's sophisticated data analysis requirements.

A data warehouse integrates data originating from multiple sources and various timeframes. It consolidates data in multidimensional space to form partially materialized data cubes. The data cube model not only facilitates OLAP in multidimensional databases but also promotes multidimensional data mining.

7.5.4 Information Retrieval

Information retrieval (IR) is the science of searching for documents or information in documents. Documents can be text or multimedia, and may reside on the Web. A text document which may involve one or multiple topics can be regarded as a mixture of multiple topic models. By integrating information retrieval models and data mining techniques, we can find the major topics in a collection of documents and for each document in the collection, the major topics involved.

7.6 Which Kinds of Applications Are Targeted?

7.6.1 Business Intelligence

Business intelligence (BI) technologies provide historical, current, and predictive views of business operations. Examples include reporting, online analytical processing, business performance management, competitive intelligence, benchmarking, and predictive analytics. Data mining is the core of business intelligence. Online analytical processing tools in business intelligence rely on data warehousing and multidimensional data mining. Classification and prediction techniques are the core of predictive analytics in business intelligence, for which there are many applications in analyzing markets, supplies, and sales.

7.6.2 Web Search Engines

A Web search engine is a specialized computer server that searches for information on the Web. The search results of a user query are often returned as a list sometimes called hits. The hits may consist of web pages, images, and other types of files. Some search engines also search and return data available in public databases or open directories.

Web search engines are essentially very large data mining applications. Various data mining techniques are used in all aspects of search engines, ranging from crawling e.g., deciding which pages should be crawled and the crawling frequencies, indexing

e.g, selecting pages to be indexed and deciding to which extent the index should be constructed, and searching e.g, deciding how pages should be ranked, which advertisements should be added, and how the search results can be personalized or made “context aware”.

Search engines pose grand challenges to data mining. First, they have to handle a huge and ever-growing amount of data. Second, Web search engines often have to deal with online data. Another challenge is maintaining and incrementally updating a model on fast growing data streams. Third, Web search engines often have to deal with queries that are asked only a very small number of times.

7.7 Major Issues in Data Mining

Major issues in data mining research are categorized into five groups: mining methodology, user interaction, efficiency and scalability, diversity of data types, and data mining and society.

7.7.1 Mining Methodology

Mining various and new kinds of knowledge: Data mining covers a wide spectrum of data analysis and knowledge discovery tasks, from data characterization and discrimination to association and correlation analysis, classification, regression, clustering, outlier analysis, sequence analysis, and trend and evolution analysis. These tasks may use the same database in different ways and require the development of numerous data mining techniques.

Mining knowledge in multidimensional space: When searching for knowledge in large data sets, we can explore the data in multidimensional space. That is, we can search for interesting patterns among combinations of dimensions (attributes) at varying levels of abstraction. Such mining is known as (exploratory) multidimensional data mining. In many cases, data can be aggregated or viewed as a multidimensional data cube. Mining knowledge in cube space can substantially enhance the power and flexibility of data mining.

Data mining an interdisciplinary effort: The power of data mining can be substantially enhanced by integrating new methods from multiple disciplines. Boosting the power of discovery in a networked environment: Most data objects reside in a linked or interconnected environment, whether it be the Web, database relations, files, or documents. Semantic links across multiple data objects can be used to advantage in data mining. Handling uncertainty, noise, or incompleteness of data: Data often contain noise, errors, exceptions, or uncertainty, or are incomplete. Errors and noise may confuse the data mining process, leading to the derivation of erroneous patterns.

Pattern evaluation and pattern- or constraint-guided mining: Not all the patterns generated by data mining processes are interesting. What makes a pattern interesting may vary from user to user. Therefore, techniques are needed to assess the interestingness of discovered patterns based on subjective measures.

7.7.2 User Interaction

The user plays an important role in the data mining process.

Interactive mining: The data mining process should be highly interactive. Thus, it is important to build flexible user interfaces and an exploratory mining environment, facilitating the user's interaction with the system. Incorporation of background knowledge: Background knowledge, constraints, rules, and other information regarding the domain under study should be incorporated into the knowledge discovery process.

Ad hoc data mining and data mining query languages: Query languages e.g, SQL have played an important role in flexible searching because they allow users to pose ad hoc queries. Similarly, high-level data mining query languages or other high-level flexible user interfaces will give users the freedom to define ad hoc data mining tasks.

Presentation and visualization of data mining results: The data mining system should present data mining results vividly and flexibly, so that the discovered knowledge can be easily understood and directly usable by humans. This is especially

data mining process is interactive. It requires the system to adopt expressive knowledge representations, user-friendly interfaces, and visualization techniques.

7.7.3 Efficiency and Scalability

Efficiency and scalability are always considered when comparing data mining algorithms. As data amounts continue to multiply, these two factors are especially critical.

Efficiency and scalability of data mining algorithms: Data mining algorithms must be efficient and scalable in order to effectively extract information from huge amounts of data in many data repositories or in dynamic data streams.

Parallel, distributed and incremental mining algorithms: The enormous size of many data sets, the wide distribution of data and the computational complexity of some data mining methods are factors that motivate the development of parallel and distributed data-intensive mining algorithms.

Cloud computing and cluster computing, which use computers in a distributed and collaborative way to tackle very large-scale computational tasks, are also active research themes in parallel data mining.

7.7.4 Diversity of Database Types

The wide diversity of database types brings about challenges to data mining.

These include Handling complex types of data: Diverse applications generate a wide spectrum of new data types, from structured data such as relational and data warehouse data to semi-structured and unstructured data, from stable data repositories to dynamic data streams, from simple data objects to temporal data, biological sequences, sensor data, spatial data, hypertext data, multimedia data, software program code, Web data and social network data.

Mining dynamic, networked, and global data repositories: Multiple sources of data are connected by the internet and various kinds of networks, forming gigantic, distributed, OF



and heterogeneous global information systems and networks. The discovery of knowledge from different sources of structured, semi-structured, or unstructured yet interconnected data with diverse data semantics poses great challenges to data mining.

Efficiency and scalability of data mining algorithms: Data mining algorithms must be efficient and scalable in order to effectively extract information from huge amounts of data in many data repositories or in dynamic data streams.

Parallel, distributed and incremental mining algorithms: The enormous size of many data sets, the wide distribution of data and the computational complexity of some data mining methods are factors that motivate the development of parallel and distributed data-intensive mining algorithms.

Cloud computing and cluster computing, which use computers in a distributed and collaborative way to tackle very large-scale computational tasks, are also active research themes in parallel data mining.

7.7.5 Data Mining and Society

Social impacts of data mining: The improper disclosure or use of data and the potential violation of individual privacy and data protection rights are areas of concern that need to be addressed. **Privacy-preserving data mining:** Data mining will help scientific discovery, business management, economy recovery and security protection e.g, the real-time discovery of intruders and cyber-attacks.

Invisible data mining: More and more systems should have data mining functions built within so that people can perform data mining or use data mining results simply by mouse clicking, without any knowledge of data mining algorithms. Intelligent search engines and Internet-based stores perform such invisible data mining by incorporating data mining into their components to improve their functionality and performance.

8. Basic Statistical Descriptions of Data

Basic statistical descriptions can be used to identify properties of the data and highlight which data values should be treated as noise or outliers.

8.1 Measuring the Central Tendency: Mean, Median, and Mode

The attribute X , like salary, is recorded for a set of objects. Let $x_1, x_2, \dots, x_N$ be the set of N observed values or observations for X . Here, these values may also be referred to as the data set (for X). To plot the observations for salary, where most of the values fall gives us an idea of the central tendency of the data. Measures of central tendency include the mean, median, mode, and midrange. The most common and effective numeric measure of the “center” of a set of data is the (arithmetic) mean. Let $x_1, x_2, \dots, x_N$ be a set of N values or observations, such as for some numeric attribute X , like salary. The mean of this set of values is

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N} = \frac{x_1 + x_2 + \dots + x_N}{N}$$

This corresponds to the built-in aggregate function, average (`avg()` in SQL), provided in relational database systems.

8.2 Measuring the Dispersion of Data: Range, Quartiles, Variance, Standard Deviation and Interquartile Range

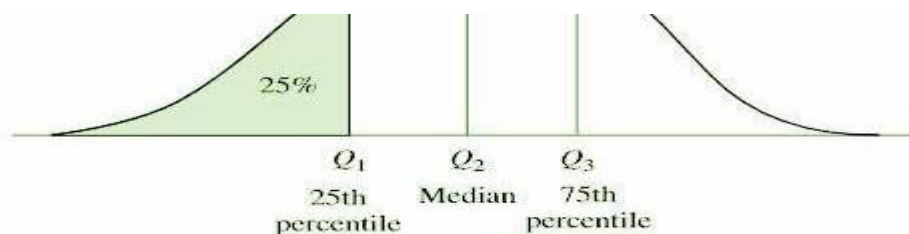
The measures such as range, quantiles, quartiles, percentiles, and the interquartile range are used to assess the dispersion or spread of numeric data. The five-number summary which can be displayed as a boxplot is useful in identifying outliers. Variance and standard deviation also indicate the spread of a data distribution.

8.2.1 Range, Quartiles and Interquartile Range

Let $x_1, x_2, \dots, x_N$ be a set of observations for some numeric attribute, X . The range of the set is the difference between the largest (`max()`) and smallest (minimum) values.

Quantiles are points taken at regular intervals of a data distribution, dividing it into essentially equal size consecutive sets. The 2-quantile is the data point dividing the lower and upper halves of the data distribution. It corresponds to the median. The 4-quantiles are the three data points that split the data distribution into four equal parts; each part represents one-fourth of the data distribution. They are more commonly referred to as quartiles.

The 100-quantiles are more commonly referred to as percentiles, they divide the data distribution into 100 equal-sized consecutive sets. The median, quartiles, and



percentiles

are

the most widely used forms of quantiles.

Figure 8.2.1 A plot of the data distribution for some attribute X.

The quantiles plotted are quartiles. The three quartiles divide the distribution into four equal-size consecutive subsets. The second quartile corresponds to the median.

The quartiles give an indication of a distribution's center, spread and shape. The first quartile, denoted by Q_1 , is the 25th percentile. It cuts off the lowest 25% of the data. The third quartile, denoted by Q_3 , is the 75th percentile. It cuts off the lowest 75% (or highest 25%) of the data. The second quartile is the 50th percentile. As the median, it gives the center of the data distribution. The distance between the first and third quartiles is a simple measure of spread that gives the range covered by the middle half of the data. This distance is called the inter quartile range (IQR) and is defined as

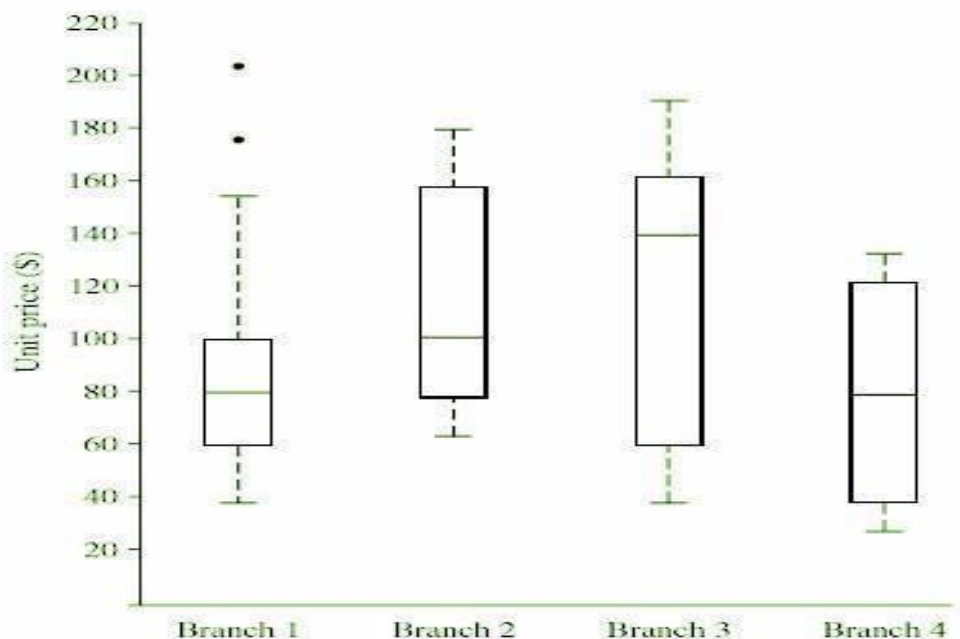
$$IQR = Q_3 - Q_1.$$

8.2.2 Five-Number Summary, Boxplots and Outliers

The five-number summary of a distribution consists of the median (Q_2), the quartiles Q_1 and Q_3 , and the smallest and largest individual observations, written in the order of Minimum, Q_1 , Median, Q_3 , and Maximum. Boxplots are a popular way of visualizing a distribution. A boxplot incorporates the five-number summary as follows:

- Typically, the ends of the box are at the quartiles so that the box length is the interquartile range.
- The median is marked by a line within the box.
- Two lines called whiskers outside the box extend to the smallest (Minimum) and largest (Maximum) observations.

Figure 2.3 Boxplot for the unit price data for items sold at four branches of All Electronics during a given time period.



8.2.3 Variance and Standard Deviation

Variance and standard deviation are measures of data dispersion. They indicate how spread out a data distribution is. A low standard deviation means that the data observations tend to be very close to the mean, while a high standard deviation indicates that the data are spread out over a large range of values.

The variance of N observations, $x_1, x_2, \dots, x_N$, for a numeric attribute X is

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 = \left(\frac{1}{N} \sum_{i=1}^N x_i^2 \right) - \bar{x}^2$$

where $\bar{x}$ is the mean value of the observations, as defined in Equation. The standard deviation, σ , of the observations is the square root of the variance, σ^2 .

8.3 Graphic Displays of Basic Statistical Descriptions of Data

Quantile plots, quantile–quantile plots, histograms, and scatter plots. These graphs are helpful for the visual inspection of data, which is useful for data preprocessing. The first three of these show univariate distributions (i.e., data for one attribute), while scatter plots show bivariate distributions (i.e., involving two attributes).

Unit – I

Video Links

Sl. No.	Topic	Video Link
1	Data Science Process	https://www.youtube.com/watch?v=s4uF8UOJz9k
2	Data Science Life Cycle	https://www.youtube.com/watch?v=4Cp6PkBKqX4
3	Exploratory Data analysis	https://www.youtube.com/watch?v=-o3AxdVcUtQ
4	Facets of Data	https://www.youtube.com/watch?v=WVclIFyCCOo
5	Data Mining	https://www.youtube.com/watch?v=0Q7j7sv4rns
6	Data Warehousing	https://www.youtube.com/watch?v=jmwGNhUXn_o
7	Statistical description of Data	https://www.youtube.com/watch?v=CVdcr_MC2KU

10. ASSIGNMENT: UNIT – I

1. Explain the data Science design process with the help of Predicting World Happiness real time application. **(CO1 , K3)**
2. Identify the applications of Data Science. **(CO1 , K2)**
3. What is the importance of Data Mining in Data Science. **(CO1 , K2)**
4. Explain how the Data Preparation work.**(CO1,K2)**
5. What are the types of data that data scientists work with to derive insights, build models, and make predictions.**(CO1,K2)**



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PART-A Q&A UNIT-1

11. PART A : Q & A : UNIT – I

1. What is Data Science ? (CO1 , K1)

As the amount of data continues to grow, the need to leverage it becomes more important. Data science involves using methods to analyze massive amounts of data and extract the knowledge it contains.

Data science is an evolutionary extension of statistics capable of dealing with the massive amounts of data produced today. It adds methods from computer science to the repertoire of statistics.

2. What is the role of data science in commercial companies? (CO1 , K1)(Gate)

Commercial companies in almost every industry use data science and big data to gain insights into their customers, processes, staff, completion, and products.

Many companies use data science to offer customers a better user experience, as well as to cross-sell, up-sell, and personalize their offerings.

Human resource professionals use people analytics and text mining to screen candidates, monitor the mood of employees, and study informal networks among coworkers.

3. What is a project charter? (CO1 , K1)

Clients like to know upfront what they are paying for, so after getting a good understanding of the business problem, try to get a formal agreement on the deliverables. All this information is collected in a project charter.

The outcome should be a clear research goal, a good understanding of the context, well-defined deliverables and a plan of action with a timetable. This information is then placed in a project charter.

4. List the steps involved in the data cleansing process. (CO1 , K2)

Errors from data entry

Physically impossible values

Missing values
Outliers
Spaces and types
Errors against codebook

5. What do you mean by outliers? (**CO1 , K2**)

An outlier is an observation that seems to be distant from other observations or, more specifically, one observation that follows a different logic or generative process than the other observations.

The easiest way to find outliers is to use a plot or a table with the minimum and maximum values.

6. What are the two operations used to combine information from different datasets? (**CO1 , K1**)

- The first operation is joining: enriching an observation from one table with information from another table.
- The second operation is appending or stacking: adding the observations of one table to those of another table.

7. What do you mean by Exploratory data analysis? (**CO1 , K2**)(Gate Question)

Information becomes much easier to grasp when shown in a picture, therefore we mainly use graphical techniques to gain an understanding of data and the interactions between variables. This phase is about exploring data, discovering anomalies missed before and to fix them.

The visualization techniques used in this phase range from simple line graphs or histograms to more complex diagrams such as Sankey and network graphs.

8. What is a pareto diagram ? (**CO1 , K1**)

We can combine simple graphs into a Pareto diagram, or 80-20 diagram. A Pareto diagram is a combination of the values and a cumulative distribution.

9. What is the use of link and brush ? (CO1 , K1)(Gate Question)

Link and brush allows us to select observations in one plot and highlight the same observations in the other plots.

With brushing and linking we combine and link different graphs and tables (or views) so changes in one graph are automatically transferred to the other graphs. This interactive exploration of data facilitates the discovery of new insights.

10. What are the steps involved in building a model? (CO1 , K2)

Building a model is an iterative process. Most of the models consist of the following main steps:

1. Selection of a modeling technique and variables to enter in the model
2. Execution of the model
3. Diagnosis and model comparison

11. What is data mining? (CO1 , K1)(Gate Question)

Data mining is searching for knowledge (interesting patterns) in data. Data mining is an essential step in the process of knowledge discovery.

Data mining provides tools to discover knowledge from data and it turns a large collection of data into knowledge.

12. What is a data warehouse? (CO1 , K1)

A data warehouse is a repository of information collected from multiple sources stored under a unified schema and usually residing at a single site.

Data warehouses are constructed via a process of data cleaning, data integration, data transformation, data loading, and periodic data refreshing.

13. What do you mean by OLAP? (CO1 , K2)

By providing multidimensional data views and the precomputation of summarized data, data warehouse systems can provide inherent support for OLAP. Online analytical processing operations make use of background knowledge regarding the

domain of the data being studied to allow the presentation of data at different levels of abstraction.

Such operations accommodate different user viewpoints. Examples of OLAP operations include drill-down and roll-up, which allow the user to view the data at differing degrees of summarization

14. What kinds of patterns can be mined? (**CO1 , K2**)(Gate Que)

- Class/Concept Description: Characterization and Discrimination
- Mining Frequent Patterns, Associations, and Correlations
- Classification and Regression for Predictive Analysis
- Cluster Analysis
- Outlier Analysis

15. What are the major issues in Data Mining? (**CO1 , K2**)

- mining methodology
- user interaction
- efficiency and scalability
- diversity of data types
- data mining and society

16. What is the need for basic statistical descriptions of data? (**CO1 , K1**)(Gate Que)

Basic statistical descriptions can be used to identify properties of the data.

It highlights which data values should be treated as noise or outliers.

17. What do you mean by interquartile range ? (**CO1 , K2**)

The distance between the first and third quartiles is a simple measure of spread that gives the range covered by the middle half of the data.

This distance is called the inter quartile range (IQR) and is defined as $IQR = Q_3 - Q_1$.

18. What is a boxplot and what do we use it ? (**CO1 , K2**)

Boxplots are a popular way of visualizing a distribution. A boxplot incorporates the five-number summary as follows:

- Typically, the ends of the box are at the quartiles so that the box length is the interquartile range.
- The median is marked by a line within the box.
- Two lines called whiskers outside the box extend to the smallest (Minimum) and largest (Maximum) observations.

19. What do you mean by Five- number summary? (**CO1 , K1**) (Gate Questions)

The five-number summary of a distribution consists of the median (Q2), the quartiles Q1 and Q3, and the smallest and largest individual observations, written in the order of Minimum, Q1, Median, Q3, and Maximum.

20. What do you mean by external data? (**CO1 , K1**)

Although data is considered an asset more valuable by certain companies, more and more governments and organizations share their data for free with the world.

This data can be of excellent quality and it depends on the institution that creates and manages it. The information they share covers a broad range of topics in a certain region and its demographics.

21. Give any two approaches to handle missing values in big data. (CO1,K2) (Gate Que)

- None: Pythonic missing data. The first sentinel value used by Pandas is None, a Python singleton object that is often used for missing data in Python code. Because it is a Python object, None cannot be used in any arbitrary NumPy/Pandas array, but only in arrays with data type 'object' (i.e., arrays of Python objects)
- NaN: Missing numerical data NaN (acronym for Not a Number) is a special floating-point value recognized by all systems that use the standard IEEE floating-point representation

12. PART B QUESTIONS: UNIT – I

1. What are the Benefits and uses of data science? (CO1 , K1)
2. Analyze the various facets of data? Explain each facet of data with examples and discuss their advantages and disadvantages. (CO1 , K3)
3. Describe the overview of the data science process. (CO1 , K2)
4. Explain in detail the data exploration, data modeling and presentation and automation steps of the data science process.(CO1,K3)
5. Explain the steps involved in the knowledge discovery process. (CO1 , K2)
6. What kinds of data can be mined? (CO1 , K2)
7. What are the technologies used in data mining? (CO1 , K2)



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13. SUPPORTIVE ONLINE CERTIFICATION COURSES

NPTEL : https://onlinecourses.nptel.ac.in/noc21_cs69/preview?

Coursera : <https://www.coursera.org/learn/python-data-analysis>

Udemy : <https://www.udemy.com/topic/data-science/>

Mooc : <https://mooc.es/course/introduction-to-datascienc>

edx : <https://learning.edx.org/course/course>



14. REAL TIME APPLICATIONS

1. Customer Exit Rate at a Bank:

Data science uses its raw data to help solve problems. A bank needed to understand why customers were leaving, this example focuses on data mining using Tableau Bank is doing a bit of data cleaning using Python. The customer loads a CSV file and discovers missing values in some subsets, such as the geography field. In this case, the data scientist needs to fill in the empty values with something to even out the data set, so the data is filled in with the "mean" score by writing a piece of code to do so. Otherwise, statistical data won't work. the data scientist uses Tableau to look at the exit rate of the bank's customers based on gender, credit card holding and geography to see if these are affecting that rate. It produces the result that the bank should consider the gender and location of its customers when analyzing how it can better retain them. Data science Plays a Major role then, the bank learns important information about client behavior.

2. Predicting World Happiness:

Predicting world happiness sounds like an impossible goal, no? Thanks to data science, it's not! Rather, using multiple linear regression model building, it's possible to assess it. To do this, one first must ascribe values. In this case, they are happiness rank, happiness value, country, region, economy, family, health, freedom, trust, generosity and dystopian residual. Not all need to be used, but some must be to make and train the model. To do this, one first must ascribe values. In this case, they are happiness rank, happiness value, country, region, economy, family, health, freedom, trust, generosity and dystopian residual. Not all need to be used, but some must be to make and train the model. As the data finishes processing, it's possible to remove the country names and plot out the most important factors that determine world happiness. The top one, as you might imagine, is the happiness score. From the analysis, the second

most important element is the economy, then family and health. Thanks to the highly detailed workings of Python's multiple linear regression model building, we can now predict world happiness.

3. Virtual Assistant:

A virtual assistant, also called an AI assistant or digital assistant, is an application program that understands natural language voice commands and completes tasks for the user. Virtual Assistants powered with AI technologies are becoming extremely popular and are taking over the world by a storm.

We have virtual assistants like Google AI, Siri, Alexa, Cortana, and many other similar virtual assistants. With the help of these assistants, we can pass commands, and using speech recognition, it tries to interpret what we are saying and automates/performs a realistic task. Using these virtual assistants, we can make calls, send messages or emails, or browse the web with just a simple voice command. We can also converse with these virtual assistants, and hence they can also act as chatbots.

15. CONTENTS BEYOND SYLLABUS: UNIT – I

VARIOUS DOMAINS OF BIG DATA ANALYTICS

Understand and Optimize Business Processes:

Big data is also increasingly used to optimize business processes. Retailers are able to optimize their stock based on predictive models generated from social media data, web search trends and weather forecasts. Another example is supply chain or delivery route optimization using data from geographic positioning and radio frequency identification sensors.

Improving Health:

The computing power of big data analytics enables us to find new cures and better understand and predict disease patterns.

We can use all the data from smart watches and wearable devices to better understand links between lifestyles and diseases.

Big data analytics also allow us to monitor and predict epidemics and disease outbreaks, simply by listening to what people are saying, i.e.

“Feeling rubbish today - in bed with a cold” or searching for on the Internet, i.e. “cures for flu”.

Improving Sports Performance:

Most elite sports have now embraced big data analytics. Many use video analytics to track the performance of every player in a football or baseball game, sensor technology is built into sports equipment such as basket balls or golf clubs, and many elite sports teams track athletes outside of the sporting environment – using smart technology to track nutrition and sleep, as well as social media conversations to monitor emotional wellbeing

Improving Security and Law Enforcement:

Security services use big data analytics to foil terrorist plots and detect cyber attacks

Police forces use big data tools to catch criminals and even predict criminal activity and credit card companies use big data analytics it to detect fraudulent transactions.

Improving and Optimizing Cities and Countries:

Big data is used to improve many aspects of our cities and countries. For example, it allows cities to optimize traffic flows based on real time traffic information as well as social media and weather data. A number of cities are currently using big data analytics with the aim of turning themselves into Smart Cities, where the transport infrastructure and utility processes are all joined up. Where a bus would wait for a delayed train and where traffic signals predict traffic volumes and operate to minimize jams.



Assessment Schedule (Proposed Date & Actual Date)

16. Assessment Schedule (Proposed Date & Actual Date)

Sl. No.	ASSESSMENT	Proposed Date	Actual Date
1	FIRST INTERNAL ASSESSMENT		
2	SECOND INTERNAL ASSESSMENT		
3	MODEL EXAMINATION		
4	END SEMESTER EXAMINATION		

17. PRESCRIBED TEXT BOOKS & REFERENCE BOOKS

TEXTBOOKS:

1. Robert S. Witte and John S. Witte, "Statistics", Eleventh Edition, Wiley Publications, 2017. (Units 2, 3)
2. Roy Jafari, Hands-On Data Preprocessing in Python: Learn how to effectively prepare data for successful data analytics, Packt Publications, First Edition, 2022. (Unit 4)
3. Marek Gagolewski, Minimalist Data Wrangling with Python, Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (CC BY-NC-ND 4.0), 2022. (Unit 5)

REFERENCES:

1. David Cielen, Arno D. B. Meysman, and Mohamed Ali, "Introducing Data Science", Manning Publications, 2016. (Unit 1)
2. Hui Lin, Ming Li, Practitioner's Guide to Data Science, 1st Edition, Chapman and Hall/CRC, 2023.
3. Jiawei Han, Micheline Kamber, Jian Pei, "Data Mining: Concepts and Techniques", 3rd Edition, Morgan Kaufmann, 2012.
4. Suresh Kumar Mukhiya, Usman Ahmed, Hands-On Exploratory Data Analysis with Python: Perform EDA techniques to understand, summarize, and investigate your data, Packt Publications, 2020.
5. Dr. Tirthajyoti Sarkar, Shubhadeep Roychowdhury, Data Wrangling with Python: Creating actionable data from raw sources, Packt Publications, First Edition, 2019.
6. Robert H. Shumway, David S. Stoffer, Time Series Analysis and Its Applications With R Examples, Fourth Edition, 2016, Springer.
7. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements Of Statistical Learning: Data Mining, Inference, and Prediction, Springer, Second Edition, 2017.
8. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, O'Reilly, Third Edition, 2022.
9. Jake VanderPlas, "Python Data Science Handbook – Essential tools for working with data", O'Reilly, 2017.
10. Ashwin Pajankar, Aditya Joshi, Hands-on Machine Learning with Python: Implement Neural Network Solutions with Scikit-learn and PyTorch, Apress, 2022.

NPTEL Courses:

11. Data Science for Engineers - https://onlinecourses.nptel.ac.in/noc23_cs17/preview
- Python for Data Science - https://onlinecourses.nptel.ac.in/noc23_cs21/preview

LIST OF EQUIPMENTS:

Systems with Anaconda, Jupyter Notebook, Python, Pandas, NumPy, Matplotlib

E-Book links:

1. <https://drive.google.com/file/d/1HoGVyZqLTQj0aA4THAD4jJ74czxEKH/view?usp=sharing>
2. <https://drive.google.com/file/d/1vJfX5xipCHZOleWfM9aUeK8mwsal6Il1/view?usp=sharing>
3. <https://drive.google.com/file/d/1aU2UKdLxLdGpmI73S1bifK8JPiMXlpoS/view?usp=sharing>

18. MINI PROJECT SUGGESTION

- a) List the Features for Student Performance Prediction and describe them in detail. How the data or feature is prepared to design model.
- b) List the Features for Sentiment Analysis For Twitter Data and describe them in detail. How the data or feature is prepared to design model.
- c) List the Features for Air Pollution Prediction and describe them in detail. How the data or feature is prepared to design model.
- d) List the Features for Credit Card Fraud Detection and describe them in detail. How the data or feature is prepared to design model.
- e) List the Features for Health care Prediction (any kind of disease detection) and describe them in detail. How the data or feature is prepared to design model.



Thank you

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